CRAFT: COMPOSITIONAL REWARD ALIGNMENT FOR TARGET-HIDDEN LOOP VERIFICATION

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ABSTRACT

Valid loop invariants are closed under conjunction: useful clauses from different responses can prove a target even when no response does so alone. Yet standard generation and training treat each response as an indivisible answer. We study this mismatch under *target-hidden loop verification*, where target assertions and postconditions $\mathcal{G}$ are withheld during generation and training. CRAFT decomposes each response into clauses, pools and filters clauses across responses, and composes the survivors before restoring $\mathcal{G}$. This preserves valid parts of imperfect responses and combines complementary parts discovered separately. Training uses the same filtered result: rejection-filtered SFT distills accepted compositions into single responses, while RL scores each response by its coverage of target-independent negative traces. Both stages transfer gains from multi-response exploration to smaller inference budgets. On 832 C programs, compositional inference raises bare Qwen3-8B from 6.43% pass@1 to 37.86% compose@1 without additional model calls. SFT reaches 63.90% compose@1. Binary and whole-response credit substantially reduce composition.

1 INTRODUCTION

Loop invariants turn unbounded executions into finite proofs, but a generated response need not be wholly correct or wholly useless. One incorrect clause can invalidate the complete response even when its other clauses are valid and useful. CRAFT separates these two cases by decomposing each response into clauses and verifying them jointly after pooling. It can then compose complementary valid clauses across responses, so even two responses that fail individually may yield a correct invariant together.

This creates an inherent tension between pass and compose. Emitting more candidate clauses increases the chance that one erroneous clause invalidates a complete response, but also enlarges the set of discoveries available to filtering and cross-response composition. CRAFT therefore does not require each raw response to succeed independently; its primary objective is the success of the filtered composition.

Whole-response training is mismatched to this inference procedure. A failed clause can obscure useful clauses from the same response. More generally, response-level RLVR can sharpen probability on already-supported answers without expanding large-k coverage (Yue et al., 2025), potentially reducing the diversity that composition needs. Our central claim is therefore that training feedback should reflect the result produced after decomposition, filtering, and composition. SFT uses accepted compositions as targets, while RL rewards the negative coverage of each response's surviving clauses.

We study this claim under *target-hidden loop verification*. The model sees the loop context but not the target assertions and postconditions $\mathcal{G}$. After the invariant set is fixed, the untouched source is restored and the verifier checks both induction and every original target. Hiding $\mathcal{G}$ prevents target copying and target-specific proof feedback, so generation and training require a graded, target-independent strength signal.

At inference, CRAFT implements *rollout–decompose–pool–filter–compose*: given k responses, it extracts their clauses, pools and filters them with syntax checks and Houdini, and composes the survivors before restoring $\mathcal{G}$. compose@k measures the verified rate of this composed result, preserving useful clauses that pass@k discards; compose@1 is our primary evaluation and reporting

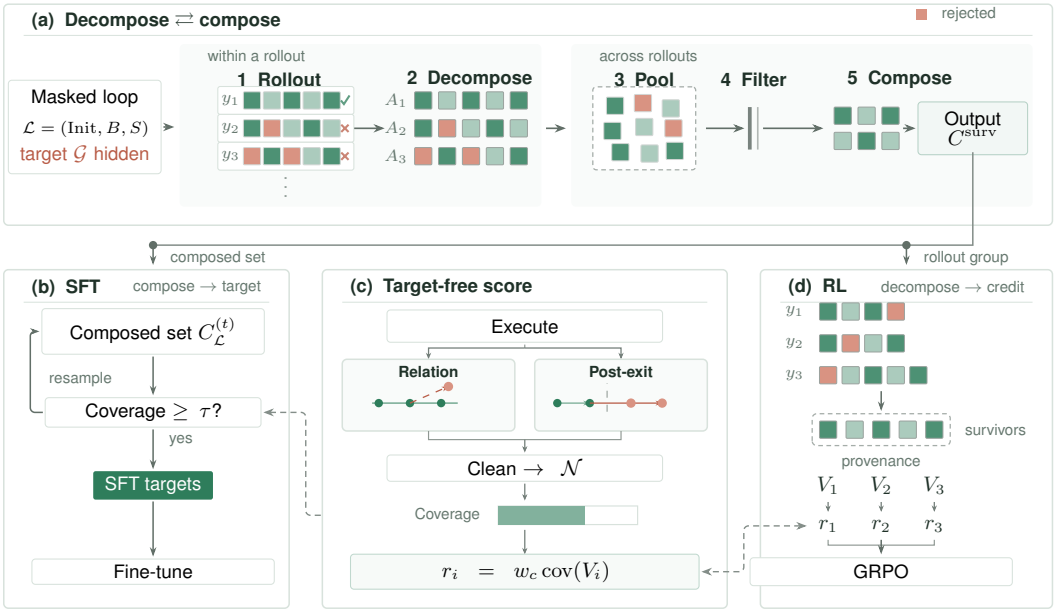

Figure 1: **Decompose–compose is the organizing principle.** Target-hidden rollouts are decomposed into clauses, so one failed clause leaves the response's other clauses intact; clauses are pooled, filtered once per group (syntax, deduplication, Houdini), and composed into one invariant (top). SFT distills an accepted multi-response pool into one response (left). Relation perturbations and post-exit continuations are cleaned and budgeted to form target-independent negative traces for coverage scoring (center; Appendix D.2); RL credits each pooled survivor to its proposer rollouts and optimizes the resulting rewards with GRPO (right). The hidden target $\mathcal{G}$ is restored only for final verification.

metric. We use accepted offline compositions to supervise SFT, then optimize the coverage of each response's surviving clauses with RL. This sequence makes discoveries obtained from multiple responses available with fewer inference samples. All supervision is self-generated: the program executor and verifier determine SFT acceptance and RL rewards, without an external teacher or reward model.

Contributions. This paper makes three contributions:

- We introduce a rollout–decompose–pool–filter–compose inference framework for target-hidden invariant generation. With one Qwen3-8B response, clause filtering and composition raise verification from 6.43% to 37.86% without another model call.
- We align SFT targets and RL credit with compositional inference. On Qwen3-8B, SFT raises compose@1 to 63.90%.
- Through controlled reward ablations, cross-model evaluation, and 832-program analysis, we demonstrate the importance of aligning training credit with the decomposed, pooled result used at inference.

2 Preliminaries

Loop verification. A task contains a loop $\mathcal{L} = (\mathsf{Init}, B, S)$ and targets $\mathcal{G} = \{(\ell, g_\ell)\}$, where Init describes loop entry, B is the guard, S is the body, and g_ℓ occurs at program location ℓ. In this paper, *valid invariant* means inductive:

$$\mathsf{Init} \Rightarrow I, \qquad \{I \wedge B\}\, S\, \{I\},$$

which implies every reachable loop-head state satisfies I (Floyd, 1967; Hoare, 1969). A finite clause set C composes as $I(C) = \bigwedge_{c \in C} c$ and is *jointly inductive* when $I(C)$ is valid. It is target-sufficient

when it proves every g_ℓ at its original location; for an immediate post-loop target with only guard-false exits and a side-effect-free guard, this reduces to $I(C) \wedge \neg B \Rightarrow g$.

Targets at different locations. Let Σ be the state space and $S \subseteq \Sigma \times \Sigma$ the relation for one normally completing loop step. Write S_ℓ^B for the path-sensitive body prefix reaching location ℓ, and X_ℓ for paths from a loop head through an exit to that location, including guard-false and `break` exits. Target sufficiency requires the corresponding obligation for every $(\ell, g) \in \mathcal{G}$:

$$\begin{aligned} I(\sigma) \wedge B(\sigma) \wedge S_\ell^B(\sigma, \sigma') &\Rightarrow g(\sigma') && \text{(body target)}, \\ I(\sigma) \wedge X_\ell(\sigma, \sigma') &\Rightarrow g(\sigma') && \text{(post-loop target)}. \end{aligned} \tag{1}$$

Both obligations quantify universally over σ, σ'. The relations include feasible-path conditions and executable side effects; X_ℓ includes body prefixes leading to `break`, not just $\neg B$; Appendix A.1 defines it explicitly. Thus joint inductiveness and target sufficiency are separate requirements: an inductive conjunction need not prove the hidden targets, and agreement with finitely many observed states does not establish induction.

Target-hidden verification. Generation and intermediate scores use $\mathcal{L}$, not $\mathcal{G}$. Targets and target-revealing text are masked, while preconditions and executable loop semantics remain visible. The invariant is fixed before the original targets are restored for final verification. We study scalar-integer C programs with one loop; Appendix C gives the scope and an example.

3 Related Work

Invariant generation and verified filtering. Classical invariant synthesis uses abstract interpretation, affine relations, templates, interpolation, syntax-guided search, or counterexamples (Cousot & Cousot, 1977; Karr, 1976; McMillan, 2003; Alur et al., 2013; Garg et al., 2014); Houdini retains an inductive subset of proposed clauses (Flanagan & Leino, 2001). Neural and LLM systems learn construction policies or combine generation with solver feedback, static analysis, filtering, and repair (Si et al., 2018; 2020; Ryan et al., 2020; Yao et al., 2020; Yu et al., 2023; Kamath et al., 2023; Chakraborty et al., 2023; Cao et al., 2025; Wu et al., 2024a; Liu et al., 2024b; Wen et al., 2024; Ma et al., 2025; Yang et al., 2026b; Wu et al., 2024b; First et al., 2023). Clause2Inv adaptively generates and combines clauses using SMT counterexamples, but its postcondition-derived feedback cannot be used under target hiding without revealing the goal (Cao et al., 2025). CRAFT instead pools clauses across independent responses with target-independent filtering and uses the composition during both inference and training.

Training from formal and behavioral feedback. Verified data and execution or prover feedback have been used to train invariant, code, and proof models (Wei et al., 2025; Pinto et al., 2026; Le et al., 2022; Yan et al., 2025). COINS and SpecRL further use positive and negative behaviors to distinguish weak valid specifications from stronger ones (Yang et al., 2026a; Huang et al., 2026). CRAFT likewise uses behavioral evidence, but derives it from target-hidden executions and applies it only after clauses survive pooled verification.

Set-level credit in verifiable-reward RL. Per-response RL can improve pass@1 without expanding large-k coverage (Yue et al., 2025; He et al., 2025). Recent objectives directly optimize pass@k or up-weight rare correct responses (Walder & Karkhanis, 2025; Chen et al., 2025; He et al., 2025). These objectives remain response-level even when they consider a set of samples. CRAFT instead decomposes responses before assigning each response the coverage of its surviving clauses.

4 Method

CRAFT applies one clause-processing pipeline to inference, SFT target construction, and RL credit assignment. Inference verifies the composition of the filtered clause union, SFT serializes an accepted composition as one response, and RL maps surviving contributions back to their proposing rollouts. Assertions and postconditions remain hidden throughout generation and training and are restored only for final verification. Appendix A gives the complete formal definition.

4.1 TARGET-HIDDEN COMPOSABLE INFERENCE

Pooling serves two purposes: independently valid clauses can jointly prove a target that none proves alone, and clauses from different responses can support one another during inductive verification. CRAFT therefore pools the decomposed clauses before filtering, rather than requiring each response to succeed independently.

For a masked loop $\mathcal{L}$, let $\mathrm{parts}(y)$ be the ordered invariant clauses extracted from response y. Let $\mathrm{Admit}(y) = \mathrm{Unique}(\mathrm{First}_{m_{\max}}(\mathrm{parts}(y)))$, where First_m keeps the first m clauses. The value of $m_{\max}$ is reported in Table 3. Algorithm 1 defines the complete inference procedure.

Algorithm 1 Rollout–decompose–pool–filter–compose

Input: task $\mathcal{T} = (\mathcal{L}, \mathcal{G})$, policy π, number of responses n
Output: composed invariant I, final verdict v

$x \leftarrow \mathcal{L}$ by masking $\mathcal{G}$ in $\mathcal{T}$ ▷ *hide target*
$y_{1:n} \sim \pi(\cdot \mid x)$ ▷ *rollout*
$A_i \leftarrow \mathrm{Admit}(y_i)$ for $i = 1, \ldots, n$ ▷ *decompose and cap*
$P \leftarrow \mathrm{Unique}(A_1 \| \cdots \| A_n)$ ▷ *pool in response order*
$C \leftarrow \mathrm{Filter}_{\mathcal{L}}(P)$ ▷ *filter*
$I \leftarrow \bigwedge_{c \in C} c$ ▷ *compose*
$v \leftarrow \mathrm{TaskVerify}(\mathcal{T}, C)$ ▷ *restore target*
return (I, v)

Joint verification and deletion. Unique removes duplicates conservatively, keeping the first occurrence in response and clause order. The filter receives this ordered family D_0. For each $c \in D$, the loop verifier checks $\mathsf{Init} \Rightarrow c$ and $\{I(D) \wedge B\}S\{c\}$, returning valid, timeout, or unknown. With per-obligation budget δ, one deletion round is

$$D_{j+1} = \left\langle c \in D_j \;\middle|\; \mathrm{LoopVerify}_{\mathcal{L},\delta}(D_j)[c] = \mathsf{valid} \right\rangle. \tag{2}$$

After each deletion, survivors are rechecked because they may have depended on removed clauses. Only a proved fixed point is returned; timeout and unknown cause deletion, and exhausting the round budget returns an empty sequence. Under a sound verifier, the resulting conjunction is inductive. Appendix A.2 gives the bounded operator, and Appendix B establishes its soundness and the candidate-relative optimality of ideal Houdini.

Why pooling precedes filtering. Let $\mathsf{H}_{\mathcal{L}}$ denote ideal Houdini with exact logical decisions. Filtering the union retains every clause obtainable by separately filtering the responses:

$$\bigcup_i \mathsf{H}_{\mathcal{L}}(A_i) \subseteq \mathsf{H}_{\mathcal{L}}\left(\bigcup_i A_i\right). \tag{3}$$

The inclusion can be strict when supporting clauses occur in different responses; its derivation is in Appendix B. Finite prover budgets can break this inclusion, so measured compose@k need not be monotonic (Appendix M).

Only after I is fixed do we restore the hidden targets $\mathcal{G}$ and run the final verification conditions at their original locations. The inference objective is therefore

$$\max_{\pi} \Pr_{y_{1:n} \sim \pi(\cdot \mid \mathcal{L})} [\mathrm{TaskVerify}((\mathcal{L}, \mathcal{G}), C(y_{1:n}))].$$

We measure compose@n on the first n saved responses per task: success is judged on the final composition, rather than on any complete response. Both complementary target proofs and cross-response inductive support can therefore contribute to gains over pass@n.

4.2 TARGET-INDEPENDENT STRENGTH SIGNAL

Inductive validity alone does not distinguish useful invariants from weak ones such as true. With targets hidden, training therefore needs a target-independent signal of how much behavior a valid

invariant excludes. We execute the masked loop to collect reachable loop-head states $\mathcal{E}$ and construct *candidate negative traces* $\mathcal{N}$. Local relational perturbations probe constraints among variables, while continuations beyond witnessed exits probe boundary constraints. Appendix D.2 gives the complete sampler, including cleaning, per-family budgets, and random fill of unused capacity.

For a verified clause set C, the state evaluator returns $\mathrm{Eval}(c, s) \in \{\mathsf{true}, \mathsf{false}, \mathsf{unknown}\}$. A negative trace is rejected only by a definite false evaluation:

$$\mathrm{rej}(C) = \{\nu \in \mathcal{N} : \exists s \in \nu, \exists c \in C, \mathrm{Eval}(c, s) = \mathsf{false}\}. \tag{4}$$

An unknown evaluation contributes no rejection. Perturbations form singleton traces, whereas a cleaned post-exit continuation remains one trace. The trajectory-level coverage is

$$\mathrm{cov}(C) = \frac{|\,\mathrm{rej}(C)|}{|\mathcal{N}|}, \qquad |\mathcal{N}| > 0. \tag{5}$$

Each trace contributes one unit regardless of its length. Instances with no retained negatives are not used for SFT acceptance; RL uses the binary all-admitted-clauses-survive fallback defined in Algorithm 4.

Empirical alignment. After inductive verification, coverage measures exclusion of sampled counterfactual behavior. Candidate negatives are not certified unreachable; the verifier supplies soundness, while coverage is an empirical strength signal rather than a guarantee of hidden-target sufficiency. On 5,711 candidate sets across 691 programs with both positive and negative final verdicts, coverage achieves a within-program macro AUROC of 0.842 (95% bootstrap CI: [0.826, 0.858]). This diagnostic uses archived target-hidden tool outputs and verified reference sets, supporting an association with final verification. Its population and limits are detailed in Appendix J.7.

4.3 Distilling Multi-Response Search

SFT aims to distill multi-response compositional search into the policy, so that a single rollout can reproduce discoveries obtained by combining multiple responses. Its target is the composed result, not a selected standalone response. It uses Algorithm 1 as a target synthesizer. Starting from an empty accumulated clause pool, each round adds a new group sampled from the same small open-weight base model that is subsequently fine-tuned. Executor-derived traces and verifier outcomes jointly determine which candidate sets become SFT targets. Each round adds the new base-model clauses and re-filters the entire accumulated pool:

$$P_t = \mathrm{Unique}\Big(P_{t-1} \| \,\mathrm{Admit}(y_1^{(t)}) \| \cdots \| \,\mathrm{Admit}(y_{g_{\mathrm{sft}}}^{(t)})\Big), \qquad C_t = \mathrm{Filter}_{\mathcal{L}}(P_t). \tag{6}$$

Accumulating raw clauses in P_t, rather than only survivors C_t, allows a removed clause to return when a later response supplies support. The target can therefore combine discoveries across responses and rounds.

Acceptance and serialization. Let t_{sft} be the synthesis round limit. For $|\mathcal{N}| > 0$, define the first accepted round and its training target by

$$\begin{aligned} t^* &= \min\{t \in \{1, \ldots, t_{\mathrm{sft}}\} : \mathrm{cov}(C_t) \geq \tau\}, \\ y^* &= \mathrm{Serialize}(C_{t^*}). \end{aligned} \tag{7}$$

If the set is empty, no training pair is produced; otherwise $(\mathcal{L}, y^*)$ enters the SFT corpus without $\mathcal{G}$. The policy is trained with standard autoregressive cross-entropy on the serialized composition y^*, conditioned on the target-masked loop. Coverage acceptance does not certify hidden-target sufficiency; Algorithm 3 gives the complete procedure.

4.4 Full decompose credit

RL aims to make useful clauses appear earlier in sampling by rewarding the parts retained after decomposition and joint filtering, even when a complete response fails. For a group sampled from the current policy, let $A_i = \mathrm{Admit}(y_i)$ as defined above. We pool and filter once, then map surviving clauses back to their proposing responses:

$$P^* = \mathrm{Unique}(A_1 \| \cdots \| A_{g_{\mathrm{rl}}}), \qquad C^* = \mathrm{Filter}_{\mathcal{L}}(P^*), \tag{8}$$

$$V_i = \{c \in A_i : \mathrm{key}(c) \in \mathrm{keys}(C^*)\}, \qquad R_i = \mathrm{rej}(V_i). \tag{9}$$

Here $\mathrm{keys}(C^*)$ is the family of canonical keys of the pooled survivors. Thus V_i assigns each pooled survivor back to every response that proposed it. A response retains the coverage contributed by its surviving clauses even when its other clauses fail, while mutually supporting clauses are checked in the same pooled context used at inference. The cheap reachable-state check is an internal front-end of this single pooled filtering call; it does not filter responses separately.

The rollout reward is

$$r_i = w_c \frac{|R_i|}{|\mathcal{N}|}. \tag{10}$$

Here w_c is the coverage weight. The per-response clause cap still limits admission, but excess clauses incur no reward penalty.

GRPO standardizes rewards within each group and applies each response's scalar advantage to its tokens. Thus clauses determine the score, while the policy update acts on the whole response rather than assigning a separate policy loss to each clause. A clause that only supports another clause receives no direct coverage credit. The update and empty-negative fallback are detailed in Appendix A.7 and Algorithm 4, respectively.

SFT learns to reproduce accepted compositions; RL rewards useful decomposed parts. Both use the pooled filtering operation applied at inference, without access to the hidden target.

5 Experiments

We ask how composition scales with sampling (RQ1), how much compositional inference and training improve verification (RQ2), how CRAFT compares with specialized tools (RQ3), how RL reshapes compositional exploration before and after SFT (RQ4), and how credit assignment affects compositional exploration under RL (RQ5).

5.1 Experimental Setup

Data and judge. Training data comprise 4,992 RL records and 33,346 SFT records (31,865 distinct sources) of target-masked scalar-integer single-loop programs; Appendix H gives their synthesis and verification. SFT targets and RL rewards are produced only by the open-weight model being trained, program execution, and Frama-C/WP verification; no external teacher or closed-model supervision is used. The evaluation combines 832 programs from Code2Inv, SyGuS, SV-COMP, LIPuS, Clause2Inv, and Loopy: 316 linear, 50 nonlinear-arithmetic, and 466 Loopy instances (Si et al., 2020; Alur et al., 2013; Beyer, 2023; Yu et al., 2023; Cao et al., 2025; Kamath et al., 2023). No training program matches an evaluation program under exact, alpha-renamed, or constant-abstracted matching (Appendix G). The judge is Frama-C 27.1 (Cobalt) with the WP plug-in, Typed memory model, and Alt-Ergo/Z3 prover portfolio (Kirchner et al., 2015). Success requires it to prove induction and every restored target; all other outcomes count as zero.

Inference protocol. All CRAFT evaluation runs share target masking, the canonical prompt, clause processing, Houdini, stochastic decoding, and no re-roll or target feedback; external tools retain their own target-hidden orchestration. RQ1 reuses one saved pool of n_{probe} responses per task. Appendix I gives decoding, serving, prover, and training details.

Metrics. Throughout evaluation, success means $\mathcal{G}$-sufficiency under the final verifier. Both metrics use the same first k saved responses per task. pass@k is the percentage of tasks for which at least one response in that prefix succeeds alone; compose@k is the percentage for which filtering and composing the prefix's clause union succeeds. Appendix I gives the definitions, and Appendix K discusses sampling variance.

5.2 RQ1: How Do pass@k and compose@k Scale?

We sample n_{probe} target-hidden responses per task from four base checkpoints: Qwen3-4B, Qwen3-8B, Qwen3-14B (Yang et al., 2025), and Llama 3.1-8B. Additional backbones and complete per-stratum results appear in Appendix J.1.

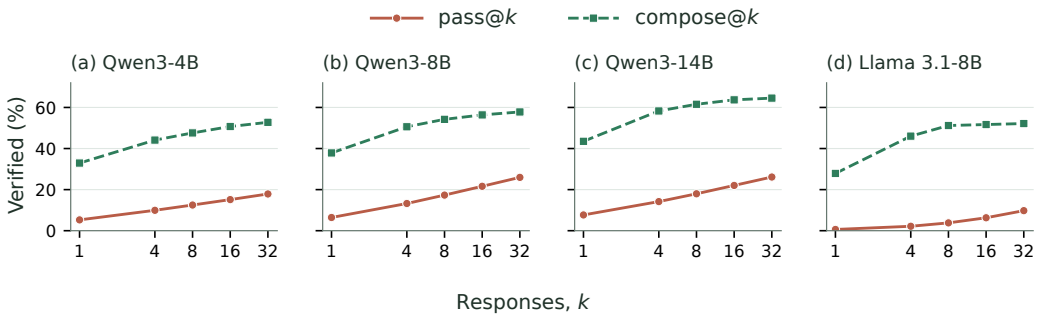

Figure 2: pass@k and compose@k on the same first k saved responses. Composition approaches saturation earlier within the tested range, while pass@k keeps growing.

At k=1, compose@1 exceeds pass@1 by 27.27–35.84 points across the four checkpoints. Both metrics evaluate the first saved response for each task: pass@1 checks its standalone success, while compose@1 decomposes and filters its clauses before final verification. Their difference therefore captures the benefit of decomposing and filtering one response, before any cross-response aggregation, subject to prefix sampling variance.

Beyond k=1, the curves scale differently. From 16 to 32 responses, compose gains at most 2.04 points across the four checkpoints, while pass gains 2.74–4.34 points. On all 832 programs, Llama 3.1-8B compose rises from 27.88% at k=1 to 51.68% at k=16, then only to 52.16% at k=32, whereas pass rises from 6.30% to 9.76% over the latter interval. We use these four backbones in the subsequent training experiments.

Finding. Under the same model-call and response budget, compose@k substantially exceeds pass@k and approaches saturation earlier within the tested range. Compositional utility should therefore be treated as a first-class optimization target rather than a by-product of standalone correctness.

5.3 RQ2: How Much Do Compositional Inference and Training Improve Verification?

For each local backbone and training stage, Table 1 reports pass@k and compose@k side by side on all 832 programs at k=1, 8, covering Bare, RL, SFT, and SFT+RL, together with four reasoning-disabled API models. Bare is the untrained checkpoint, and RL applies reinforcement learning directly to Bare. Complete API results at $k \in \{1, 4, 8\}$ are in Table 19. Across all four local backbones, SFT raises compose@1 to 61.30–66.35%. Further RL raises Qwen3-8B compose@1 from 63.90% to 69.23%, an additional 5.33 percentage points. Section 5.5 compares the effects of RL from Bare and SFT initializations. Full sampling curves and benchmark-stratum breakdowns are deferred to Appendix J.

On Qwen3-8B, one SFT rollout reaches 63.90% compose@1, exceeding the Bare model's 57.81% compose@32 by 6.09 percentage points (Table 18). Thus SFT attains a higher verification rate with one online model call instead of 32. This comparison concerns inference calls, not total compute including offline synthesis and training.

Finding. SFT amortizes multi-response compositional search into a single rollout, and RL further improves sampling efficiency by rewarding surviving clauses. On Qwen3-8B, SFT compose@1 already exceeds Bare compose@32; further RL raises compose@1 to 69.23%.

Backbone	Training	pass@1	compose@1	pass@8	compose@8
Qwen3-4B	Bare	5.27	32.93	12.50	47.60
	RL	1.70	48.80	4.50	62.50
	SFT	40.43	62.62	59.99	76.20
	SFT+RL (CRAFT)	33.40	66.10	49.90	74.50
Qwen3-8B	Bare	6.43	37.86	17.33	54.21
	RL	6.80	57.93	16.80	70.43
	SFT	41.93	63.90	61.11	76.20
	SFT+RL (CRAFT)	30.78	69.23	47.47	76.80
Qwen3-14B	Bare	7.67	43.51	17.97	61.54
	RL	10.40	65.40	18.70	72.20
	SFT	**45.66**	66.35	**63.50**	**78.37**
	SFT+RL (CRAFT)	40.70	**70.80**	56.80	76.80
Llama 3.1-8B	Bare	0.61	27.88	3.81	51.20
	RL	4.50	46.00	10.50	56.20
	SFT	39.24	61.30	56.86	73.08
	SFT+RL (CRAFT)	33.70	63.70	49.64	72.70
API models					
GPT-5-nano	–	3.58	33.89	16.05	55.53
GPT-5	–	32.58	51.32	51.74	72.00
Claude Sonnet-4.6	–	36.35	56.97	44.93	66.95
DeepSeek V4-Flash	–	28.08	51.44	48.21	74.40

Table 1: Main results on all 832 programs (% verified). Pass evaluates complete responses, whereas compose decomposes, pools, and filters the same response pool. Bold marks the best value in each column across all rows, including the API baselines: trained local models lead every column, and CRAFT attains the best compose@1 (70.80%). API rows are evaluated without additional training. Complete curves and stratum-level results are in Appendix J.

5.4 RQ3: How Does CRAFT Compare with Specialized Tools?

We compare CRAFT with the target-hidden pipelines of AutoSpec, SESpec, Clause2Inv, Loopy, and Daikon (Wen et al., 2024; Yang et al., 2026b; Cao et al., 2025; Kamath et al., 2023; Ernst et al., 2007). The LLM-based baselines and the model-matched CRAFT runs use GPT-5-nano with reasoning disabled; the trained CRAFT checkpoint is shown as an additional series. All outputs face the same untouched Frama-C/WP judge, and unsupported inputs count as failures among all 832 programs. Figure 3 compares verification rate against mean total tokens and end-to-end time under each system's native orchestration. In the token panel, all three GPT-5-nano CRAFT budgets lie on the model-matched Pareto frontier: compose@4 verifies 49.04% with fewer tokens than AutoSpec, the strongest external baseline in this reasoning-disabled comparison (42.19%), while compose@8 reaches 55.53%.

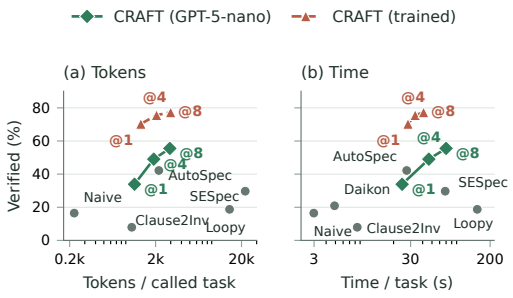

Figure 3: Verification rate versus (a) token use and (b) end-to-end time; Costs are means; upper-left is better. LLM-based runs use GPT-5-nano except CRAFT (trained). Daikon makes no model call and appears only in the time panel.

The trained CRAFT checkpoint reaches 70.31%, 75.60%, and 77.28% at compose@1, @4, and @8, using 1,348, 2,053, and 2,993 tokens per task, respectively. The corresponding mean end-to-end times are 28.2, 33.5, and 40.9 seconds per task. At compose@4 and compose@8, the trained checkpoint has both higher verification rates and lower recorded times than GPT-5-nano CRAFT (46.40 and 69.99 seconds). Wall times reflect each system's serving setup; GPT-5-nano CRAFT times combine archived request latency with measured prefix filtering and final verification.

We compare systems end to end without matching calls or token budgets. Appendix J reports accuracy by category, token use, runtime, and reasoning effort. The benefit also holds for three additional API models (Table 19).

Finding. Under target hiding, orchestration efficiency matters alongside proposal strength. CRAFT turns independent responses into reusable clauses: all three GPT-5-nano budgets lie on the model-matched token frontier, and compose@4 simultaneously beats the strongest external baseline in verification rate and mean tokens.

5.5 RQ4: How Does RL Reshape Compositional Exploration Before and After SFT?

Prior work suggests that RLVR can reallocate probability toward behaviors already available to the base model rather than create fundamentally new reasoning patterns (Yue et al., 2025). We test this at the level of compositional utility using matched Bare–RL and SFT–SFT+RL Qwen3-8B pairs (Figure 5).

With Full decompose credit, RL from Bare raises compose@1 from 37.86% to 57.93% (+20.07 points), whereas compose@32 rises from 57.81% to 73.20% (+15.39 points). After SFT, RL raises compose@1 from 63.90% to 69.23% (+5.33 points), while compose@32 changes from 77.90% to 78.37% (+0.47 points; Table 18). These results are consistent with RL primarily increasing the sampling probability of existing composable clauses rather than creating new reasoning patterns, especially after SFT initialization. After SFT, the substantial compose@1 gain alongside the small compose@32 gain suggests that useful clauses become available earlier in sampling. From Bare, the larger gain at compose@32 is also compatible with making previously rare capabilities more accessible; the finite-budget curves alone do not distinguish this explanation from the emergence of new reasoning patterns.

Meanwhile, after SFT, pass@1 falls from 41.93% to 30.78%. This is compatible with rewarding useful surviving clauses rather than requiring their entire response to succeed independently.

Finding. RL makes useful clauses available with fewer samples: its composition gains are larger at one response than at 32, both before and after SFT.

5.6 RQ5: How Should Credit Be Assigned for Compositional Inference?

Because targets remain hidden during RL, we use negative coverage of the pooled survivors as a target-independent surrogate. From the same Qwen3-8B Bare initialization, Binary and Whole-rollout use response-level credit, Full decompose credit preserves credit for surviving V_i (Equation (10)); all other settings are fixed.

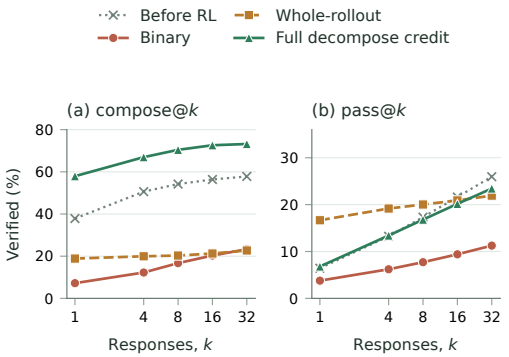

Figure 4: Qwen3-8B reward ablations from the Bare initialization. Whole-response credit reduces composition; clause credit recovers it. The matched SFT-initialized controls, where Binary collapses the strong SFT composition curve, are in Table 18.

Figure 5: Matched Qwen3-8B composition curves before and after RL. RL improves Bare throughout and raises small-k composition after SFT saturates at large k.

Binary reaches only 23.32% compose@32, below Bare's 57.81%, showing that rewarding inductive validity alone is insufficient to preserve compositional performance in this setting. Whole-rollout adds a strength signal but retains the whole-response correctness gate; comparing it with Full decompose credit isolates the effect of that gate.

Whole-rollout credit raises pass@1 from 6.43% to 16.67% yet reduces compose@32 from 57.81% to 22.72%. This exposes a tension between single-response correctness and exploration breadth. Its all-or-nothing gate withholds credit from every useful surviving clause whenever any clause in the response is rejected. The observed composition loss is consistent with suppressing useful partial candidates that pooled inference can exploit. Full decompose credit removes the gate and credits each response for the coverage of its surviving V_i. Relative to Whole-rollout, it raises compose@1 from 18.87% to 57.93% and compose@32 from 22.72% to 73.20%, surpassing Bare by 20.07 and 15.39 points, respectively. The available SFT-initialized and cross-backbone controls are reported in Table 18.

Finding. Training credit should reflect the clauses retained by compositional inference. Full decompose credit preserves useful contributions lost under Whole-rollout's all-or-nothing reward.

6 Conclusion and Limitations

Useful invariant clauses need not occur in one successful response. Cross-response filtering substantially improves bare-model verification, while SFT and RL make these gains available at smaller sampling budgets. Reward ablations show that higher pass need not imply better composition: credit aligned with the inference algebra preserves gains lost under whole-response credit. Both metrics use one saved response prefix per task and budget, so sampling variance can affect measured gaps and saturation trends. Future work should quantify this uncertainty with repeated matched sampling and test transfer to other tasks with independently verifiable components (Appendix K).

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AI Use Statement

Generative AI tools are integral to the evaluated system: they produce the invariant candidates and synthetic SFT data described in the paper. They were also used during research to refine hypotheses, methodology, and experiments; formulate and check mathematical claims; implement and test software; analyze and interpret experimental artifacts; prepare figures and tables; search and summarize literature; and draft, edit, and translate manuscript text. The authors reviewed all AI-assisted work and tested generated code. They take responsibility for independently validating the final content, empirical claims, and released artifacts.

Reproducibility Statement

The method and objective are specified in Section 4; Appendices D–I give the implementation defaults, prompt, training interface, data-overlap audit, and evaluation protocol. Appendix J organizes the available measurements.

A Complete Formalization of CRAFT

This section expands the method in Section 4 into a complete input-to-output specification. Given a task with hidden goals, CRAFT masks those goals and applies the same rollout–decompose–pool–filter–compose operator at inference time and during target construction. We first define the program semantics and verifier, then the exact and resource-bounded filters, and finally the inference, SFT, and RL procedures. Theoretical properties are stated and proved separately in Appendix B.

A.1 Task Semantics, Loop Invariants, and Verification

Let Σ be the program-state space. A loop is $\mathcal{L} = (\mathsf{Init}, B, S)$, where Init characterizes the first loop-head states, B is the guard, and $S \subseteq \Sigma \times \Sigma$ is one normally completing loop step. Thus $S(\sigma, \sigma')$ includes execution of the body and any guard or update side effects required to reach the next loop head.

Definition 1 (Inductive invariant)**.** *A predicate I is a valid loop invariant when*

$$\begin{aligned} &\forall \sigma \in \Sigma :\ \mathsf{Init}(\sigma) \Rightarrow I(\sigma), \\ &\forall \sigma, \sigma' \in \Sigma :\ I(\sigma) \wedge B(\sigma) \wedge S(\sigma, \sigma') \Rightarrow I(\sigma'). \end{aligned} \tag{11}$$

We denote these two obligations by $\mathrm{Ind}_{\mathcal{L}}(I)$*. Here and throughout the paper,* valid invariant *means inductive in this sense. This implies validity on every reachable loop-head state, although the converse need not hold for the chosen assertion language and prover.*

A response is decomposed into a finite candidate family C of clauses. The clauses compose by conjunction,

$$I(C) = \bigwedge_{c \in C} c, \qquad I(\varnothing) = \mathit{true}. \tag{12}$$

The set C is *jointly inductive* when $\mathrm{Ind}_{\mathcal{L}}(I(C))$ holds. Notice that preservation is checked under the full conjunction: a clause may rely on companion clauses. The logical definition is insensitive to clause order. The bounded verifier, however, receives a deterministic sequence $D = \langle d_1, \ldots, d_m \rangle$, because generated proof dependencies can follow annotation order. Set notation below denotes the members of this sequence; deletion preserves their relative order.

A complete task is $\mathcal{T} = (\mathcal{L}, \mathcal{G})$, where $\mathcal{G}$ contains the assertions and postconditions at their original program locations. For a body target, let $S^B_\ell \subseteq \Sigma \times \Sigma$ be the path-sensitive prefix relation from the loop head through guard evaluation to ℓ. For a target after the loop, let E_0 be the guard-false exit relation and E_b the relation from a loop head through the body prefix to a `break` that exits this loop. These relations include guard and branch conditions and all side effects. Let $F_{e,\ell}$ be the suffix from exit e to ℓ, the identity for an immediately following target. Define the complete exit-to-target relation by

$$X_\ell(\sigma, \sigma') \iff \bigvee_{e \in \{0\} \cup \mathcal{B}} \exists \eta :\ E_e(\sigma, \eta) \wedge F_{e,\ell}(\eta, \sigma'),$$

where $\mathcal{B}$ indexes the loop's break exits. For a side-effect-free guard, $E_0(\sigma, \eta)$ is $\neg B(\sigma) \wedge \eta = \sigma$. Unlike E_0, a break exit may start with $B(\sigma)$ true. Then define

$$\mathrm{VC}_{\mathcal{L}}(I, \ell, g) = \begin{cases} \forall \sigma, \sigma' : I(\sigma) \wedge B(\sigma) \wedge S^B_\ell(\sigma, \sigma') \Rightarrow g(\sigma'), & \ell \text{ is inside the body,} \\ \forall \sigma, \sigma' : I(\sigma) \wedge X_\ell(\sigma, \sigma') \Rightarrow g(\sigma'), & \ell \text{ is after the loop.} \end{cases} \tag{13}$$

Postconditions are represented by the second case, with suffix relations including assignments and return-value substitution. A return or other abrupt exit inside the body, when admitted, requires its own exit relation and applicable target obligations in the same union; it does not contribute to the normally completing transition S. For example, with `while (1) { if (q) break; ... }`, the guard-false branch is empty, but the break branch still yields a nonvacuous post-loop obligation. The ideal logical verdict is

$$\mathrm{TaskValid}_{\models}(\mathcal{T}, C) = \mathrm{Ind}_{\mathcal{L}}(I(C)) \wedge \bigwedge_{(\ell, g) \in \mathcal{G}} \mathrm{VC}_{\mathcal{L}}(I(C), \ell, g). \tag{14}$$

Generation and training see only $\mathcal{L}$; $\mathcal{G}$ is restored only for Equation (14).

For a proof goal g generated by the configured loop verifier, define its portfolio status by

$$\mathrm{GoalStatus}_\delta(g) \in \{\mathsf{valid}, \mathsf{timeout}, \mathsf{unknown}\}, \tag{15}$$

where valid means that at least one configured prover returned *Valid*; timeout means that no prover proved the goal and at least one exhausted δ; and unknown contains every other non-valid, unsupported, missing, or malformed result.

Let $\mathcal{S}_{\mathrm{ver}} = \{\mathsf{valid}, \mathsf{timeout}, \mathsf{unknown}\}$. One invocation of the resource-bounded loop verifier on the current ordered family $D = \langle d_1, \ldots, d_m \rangle$ returns the clause-to-status map

$$\mathrm{LoopVerify}_{\mathcal{L},\delta}(D) = \{d_k \mapsto v_{D,k} : 1 \le k \le m\} \in \mathcal{S}_{\mathrm{ver}}^{D}, \tag{16}$$

where $v_{D,k} = \mathsf{valid}$ means that the verifier proves all establishment and preservation obligations generated for d_k in the current family D. If the obligations are not all proved, $v_{D,k} = \mathsf{timeout}$ records an exhausted prover budget and $v_{D,k} = \mathsf{unknown}$ records every other outcome. The interface is sound when

$$\big(\forall c \in D :\ \mathrm{LoopVerify}_{\mathcal{L},\delta}(D)[c] = \mathsf{valid}\big) \Longrightarrow \mathrm{Ind}_{\mathcal{L}}(I(D)). \tag{17}$$

The filter treats timeout and unknown identically, but the interface keeps them distinct. Appendix D.1 describes how the implementation generates and aggregates the clause goals.

After the invariant set is fixed, the separate binary operator $\mathrm{TaskVerify}_\delta(\mathcal{T}, C)$ runs the verifier on the original task annotated with C. Let $\mathrm{TargetGoals}(\mathcal{T}, C)$ be its generated source-level assertion and postcondition goals. Then

$$\begin{aligned}\mathrm{TaskVerify}_\delta(\mathcal{T}, C) = \mathbf{1}\big[\,&\mathrm{SyntaxOK}(\mathcal{T}, C) \wedge \\ &\forall c \in C :\ \mathrm{LoopVerify}_{\mathcal{L},\delta}(C)[c] = \mathsf{valid} \wedge \\ &\forall g \in \mathrm{TargetGoals}(\mathcal{T}, C) : \mathrm{GoalStatus}_\delta(g) = \mathsf{valid}\big].\end{aligned} \tag{18}$$

A.2 Ideal and Resource-Bounded Houdini Filtering

We define an exact semantic fixed-point operator and its resource-bounded verifier counterpart.

For $c \in C$, define its establishment and context-dependent preservation obligations by

$$\begin{aligned}\mathrm{Est}_{\mathcal{L}}(c) &\iff \forall \sigma :\ \mathsf{Init}(\sigma) \Rightarrow c(\sigma), \\ \mathrm{Pres}_{\mathcal{L},C}(c) &\iff \forall \sigma, \sigma' : I(C)(\sigma) \wedge B(\sigma) \wedge S(\sigma, \sigma') \Rightarrow c(\sigma').\end{aligned} \tag{19}$$

Ideal Houdini starts with $C_0 = C$ and repeats

$$C_{j+1} = \mathsf{F}_{\mathcal{L}}(C_j) = \{c \in C_j : \mathrm{Est}_{\mathcal{L}}(c) \wedge \mathrm{Pres}_{\mathcal{L},C_j}(c)\} \tag{20}$$

until a fixed point $\mathsf{H}_{\mathcal{L}}(C)$ is reached.

The implemented deletion round uses the clause-level loop verifier and preserves the relative order of retained clauses:

$$\widehat{\mathsf{F}}_{\mathcal{L},\delta}(D) = \big\langle c \in D \;\big|\; \mathrm{LoopVerify}_{\mathcal{L},\delta}(D)[c] = \mathsf{valid} \big\rangle. \tag{21}$$

Thus timeout and unknown both cause conservative deletion. Every status in one round is computed under the same pre-deletion context D. A clause removed at the end of that round may therefore have helped another clause receive valid. The survivors must be verified again without the removed hypotheses; this is why a single deletion pass is insufficient. For an ordered input sequence D, let $D_0 = D$ and $D_{j+1} = \widehat{\mathsf{F}}_{\mathcal{L},\delta}(D_j)$. Iteration stops at the first proved fixed point or at the symbolic round limit $h_{\max}$. Let

$$j^* = \min\{j < h_{\max} : \widehat{\mathsf{F}}_{\mathcal{L},\delta}(D_j) = D_j\}, \tag{22}$$

when this set is nonempty. The implemented filter is

$$\mathrm{Filter}_{\mathcal{L}}^{\delta,h_{\max}}(D) = \begin{cases} D_{j^*}, & j^* \text{ exists}, \\ \langle\rangle, & \text{the round limit is reached first.} \end{cases} \tag{23}$$

At a returned fixed point, every surviving clause has just been proved in the context of that same sequence. Reaching the limit returns the empty sequence. Under a sound loop verifier, the result is sound and may be smaller than the ideal fixed point. In the remainder, $\mathrm{Filter}_{\mathcal{L}}$ abbreviates Equation (23), and TaskVerify abbreviates $\mathrm{TaskVerify}_\delta$.

A.3 The Shared Composition Operator

Let $\mathrm{parts}(y)$ extract the ordered sequence of invariant clauses in response y, let $\mathrm{First}_m(L)$ be the prefix of sequence L containing at most m clauses, and let $\|$ denote sequence concatenation. Define the clauses admitted from one response by

$$\mathrm{Admit}(y) = \mathrm{Unique}(\mathrm{First}_{m_{\max}}(\mathrm{parts}(y)))\,. \tag{24}$$

Unique removes duplicates using the finite-rule canonical normalization $\mathrm{key}(c)$. Equal keys identify the clauses merged by those rules. Provenance is retained by recording which responses supplied each key. It keeps the first occurrence; concatenations are traversed in response order and then clause order, which fixes the sequence supplied to LoopVerify. For a clause sequence D, let $\mathrm{keys}(D) = \langle \mathrm{key}(c) : c \in D \rangle$.

For any response sequence $y_{1:n}$, define

$$A_i = \mathrm{Admit}(y_i), \qquad P(y_{1:n}) = \mathrm{Unique}(A_1 \| \cdots \| A_n)\,, \tag{25}$$

$$C(y_{1:n}) = \mathrm{Filter}_{\mathcal{L}}(P(y_{1:n})), \qquad I(y_{1:n}) = \bigwedge_{c \in C(y_{1:n})} c. \tag{26}$$

This is the shared rollout–decompose–pool–filter–compose operator. The filter checks preservation under the complete current conjunction, so clauses from different responses may support one another; standalone preservation is not required.

A.4 Complete Inference Algorithm

Algorithm 2 CRAFT inference

Input: task $\mathcal{T} = (\mathcal{L}, \mathcal{G})$, policy π, number of responses n
Output: composed invariant I, final verdict v

$x \leftarrow \mathrm{Mask}(\mathcal{T}) = \mathcal{L}$ ▷ *hide goals*
sample $y_{1:n} \sim \pi(\cdot \mid x)$ ▷ *rollout*
$A_i \leftarrow \mathrm{Admit}(y_i)$ for every i ▷ *decompose and cap*
$P \leftarrow \mathrm{Unique}(A_1 \| \cdots \| A_n)$ ▷ *pool*
$C \leftarrow \mathrm{Filter}_{\mathcal{L}}^{\delta, h_{\max}}(P)$ ▷ *filter*
$I \leftarrow \bigwedge_{c \in C} c$ ▷ *compose*
$v \leftarrow \mathrm{TaskVerify}_{\delta}(\mathcal{T}, C)$ ▷ *restore goals*
return (I, v)

For evaluation, compose@n is the percentage of tasks for which $\mathrm{TaskVerify}(\mathcal{T}, C(y_{1:n}))$ succeeds on the first n saved responses. Pass@n uses that same prefix and counts a task as solved when at least one response succeeds alone (Appendix I).

A.5 Complete SFT Target Construction

SFT distills the same multi-response map into one response. It accumulates raw candidates so that a clause discarded in one round can later acquire a supporting companion. The policy π_0 is the small open-weight model being trained and is the only learned generator in target construction. Executions of $\mathcal{L}$ produce the trace signal, and $\mathrm{Filter}_{\mathcal{L}}$ supplies the verifier decision; no external model supplies candidates, labels, or acceptance judgments.

Algorithm 3 Multi-response SFT synthesis

Input: masked loop $\mathcal{L}$, policy π_0, negatives $\mathcal{N}$, group size g_{sft}, round limit t_{sft}
Output: one serialized target, or reject

if $\mathcal{N} = \varnothing$ **then return** reject
$P \leftarrow \langle\rangle$
for $t = 1, \ldots, t_{\mathrm{sft}}$ **do**
  sample a response group $y_{1:g_{\mathrm{sft}}} \sim \pi_0(\cdot \mid \mathcal{L})$
  $P \leftarrow \mathrm{Unique}(P \| \mathrm{Admit}(y_1) \| \cdots \| \mathrm{Admit}(y_{g_{\mathrm{sft}}}))$
  $C \leftarrow \mathrm{Filter}_{\mathcal{L}}(P)$
  if $\mathrm{cov}_{\mathcal{N}}(C) \geq \tau$ **then return** $\mathrm{Serialize}(C)$
return reject

The SFT input and target contain no element of $\mathcal{G}$. The parameter values appear in Appendix D.

A.6 Complete RL Group-Scoring Algorithm

The sampled policy is the same small open-weight model updated by RL. Given its rollouts, every reward component below is computed from program executions, verifier outcomes, and deterministic clause provenance; no external model enters the scoring function.

Executions of the masked loop yield reachable loop-head states $\mathcal{E}$ and target-independent negative traces $\mathcal{N}$. The lightweight evaluator is three-valued:

$$\mathrm{Eval}(c, s) \in \{\mathsf{true}, \mathsf{false}, \mathsf{unknown}\}. \tag{27}$$

An unknown result neither rejects a negative trace nor removes a candidate in the reachable-state front end; syntax and induction are left to the verifier. A trace counts as rejected only when at least one retained clause evaluates definitively to false at one of its states:

$$\mathrm{rej}_{\mathcal{N}}(C) = \{\nu \in \mathcal{N} : \exists s \in \nu, \exists c \in C,\ \mathrm{Eval}(c, s) = \mathsf{false}\}, \qquad \mathrm{cov}_{\mathcal{N}}(C) = \frac{|\,\mathrm{rej}_{\mathcal{N}}(C)|}{|\mathcal{N}|}. \tag{28}$$

Coverage is trajectory level and is defined only for $|\mathcal{N}| > 0$. Instances without retained negatives are rejected during SFT construction. For RL, the explicit fallback below uses a binary survivor signal and never evaluates this ratio.

For an RL group $y_{1:g_{\mathrm{rl}}}$, first form the same globally deduplicated pool used at inference, and then invoke the verifier filter once:

$$A_i = \mathrm{Admit}(y_i), \qquad P^* = \mathrm{Unique}(A_1 \| \cdots \| A_{g_{\mathrm{rl}}})\,, \tag{29}$$

$$C^* = \mathrm{Filter}_{\mathcal{L}}(P^*), \qquad V_i = \langle c \in A_i \mid \mathrm{key}(c) \in \mathrm{keys}(C^*)\rangle\,, \qquad R_i = \mathrm{rej}_{\mathcal{N}}(V_i). \tag{30}$$

Here V_i assigns each pooled survivor back to every response that proposed it. A reachable-state front end evaluates the globally pooled candidates on $\mathcal{E}$ before the verifier processes the same pool. The reward is

$$r_i = w_c \frac{|R_i|}{|\mathcal{N}|}. \tag{31}$$

Here w_c is the coverage weight. The reward scores each response by the negative coverage of its pooled survivors. A purely supporting clause may enable another clause to survive but receives no direct coverage credit. The per-response cap limits admitted clauses without penalizing excess output.

Algorithm 4 CRAFT reward for one RL group

Input: masked loop $\mathcal{L}$, responses $y_{1:g_{\mathrm{rl}}}$, reachable states $\mathcal{E}$, negative traces $\mathcal{N}$
Output: response rewards $r_{1:g_{\mathrm{rl}}}$

$A_i \leftarrow \mathrm{Admit}(y_i)$ for every i
$P^* \leftarrow \mathrm{Unique}(A_1 \| \cdots \| A_{g_{\mathrm{rl}}})$ ▷ *pool globally*
$C^* \leftarrow \mathrm{Filter}_{\mathcal{L}}(P^*)$ ▷ *filter the pool once*
$V_i \leftarrow \langle c \in A_i \mid \mathrm{key}(c) \in \mathrm{keys}(C^*) \rangle$ for every i ▷ *project by provenance*
if $\mathcal{N} = \varnothing$ **then**
  $r_i \leftarrow \mathbf{1}[A_i \neq \langle\rangle \ \wedge \ \mathrm{keys}(V_i) = \mathrm{keys}(A_i)]$ for every i
  return $r_{1:g_{\mathrm{rl}}}$
$R_i \leftarrow \mathrm{rej}_{\mathcal{N}}(V_i)$ for every i
$r_i \leftarrow w_c |R_i| / |\mathcal{N}|$ for every i
return $r_{1:g_{\mathrm{rl}}}$

A clause c *semantically supports* d in D when $c \neq d$, $\mathrm{Pres}_{\mathcal{L},D}(d)$ holds, but $\mathrm{Pres}_{\mathcal{L},D\setminus\{c\}}(d)$ does not. The implemented verifier exposes the broader, resource-dependent relation

$$\begin{aligned}\mathrm{VSupport}_{\mathcal{L},\delta}(c, d \mid D) \iff\ & c \neq d \ \wedge \mathrm{LoopVerify}_{\mathcal{L},\delta}(D)[d] = \mathsf{valid} \ \wedge \\ & \mathrm{LoopVerify}_{\mathcal{L},\delta}(D \setminus \{c\})[d] \neq \mathsf{valid}.\end{aligned} \tag{32}$$

This relation includes semantic support, generated-goal dependencies, and resource effects. Algorithm 4 can retain such a clause through joint filtering, but assigns it direct reward only if its projected response rejects a negative trace.

A.7 Group-Relative Policy Update

The rollout policy $\pi_{\theta_{\mathrm{old}}}$ samples a group, which is scored by Algorithm 4. Define

$$\widehat{A}_i = \frac{r_i - \mathrm{mean}_j\, r_j}{\mathrm{std}_j\, r_j + \delta_{\mathrm{adv}}}. \tag{33}$$

For token $a_{i,u}$ and its prefix $h_{i,u} = (\mathcal{L}, a_{i,<u})$, let

$$\rho_{i,u}(\theta) = \frac{\pi_\theta(a_{i,u} \mid h_{i,u})}{\pi_{\theta_{\mathrm{old}}}(a_{i,u} \mid h_{i,u})}, \tag{34}$$

$$q_{i,u}(\theta) = \frac{\pi_{\mathrm{ref}}(a_{i,u} \mid h_{i,u})}{\pi_\theta(a_{i,u} \mid h_{i,u})}, \qquad k_{i,u}(\theta) = q_{i,u} - \log q_{i,u} - 1. \tag{35}$$

The nonnegative quantity $k_{i,u}$ is the sampled conditional-token KL surrogate used by the implementation. Its expectation equals $D_{\mathrm{KL}}(\pi_\theta(\cdot \mid h_{i,u}) \| \pi_{\mathrm{ref}}(\cdot \mid h_{i,u}))$ when the token is sampled from $\pi_\theta(\cdot \mid h_{i,u})$. During an update, however, the saved token comes from $\pi_{\theta_{\mathrm{old}}}$, so $k_{i,u}$ is used as a sampled penalty rather than claimed to be an unbiased off-policy estimate. The clipped group objective is

$$\mathcal{J}(\theta) = \mathbb{E}\left[\frac{1}{g_{\mathrm{rl}}} \sum_i \frac{1}{T_i} \sum_{u=1}^{T_i} \left(\min\left\{ \rho_{i,u}\widehat{A}_i,\ \mathrm{clip}(\rho_{i,u}, 1-\varepsilon, 1+\varepsilon)\widehat{A}_i \right\} - \beta k_{i,u} \right)\right]. \tag{36}$$

After optimizing this objective, the updated policy samples the next group.

A.8 Training–Inference Alignment

Inference, SFT target construction, and RL group scoring all call the same pooled map in Equation (26); they differ only in how the returned clauses are consumed. Inference checks the restored goals, SFT serializes an accepted fixed point, and RL projects the fixed point back to the responses that proposed its surviving clauses. Negative coverage supplies the target-independent training surrogate; restored-goal verification supplies the final verdict.

A.9 Extension to Verifiably Composable Generation

Definition 2 (Verifiably composable generation)**.** *A task provides visible input x, hidden goals $\mathcal{G}$, an atomic decomposition* parts_x*, a duplicate-removal operator* Unique_x*, a subset filter* Filter_x*, a composition operator* $\mathrm{Compose}_x$*, and a validity predicate* Valid_x*. For responses $y_{1:n}$, define*

$$P_x(y_{1:n}) = \mathrm{Unique}_x\left(\bigcup_i \mathrm{parts}_x(y_i)\right), \tag{37}$$

$$O_x(y_{1:n}) = \mathrm{Compose}_x(\mathrm{Filter}_x(P_x(y_{1:n}))). \tag{38}$$

The task is verifiably composable when the filter is conservative and sound: it returns atoms from its input and $\mathrm{Valid}_x(O_x(y_{1:n}))$ *holds whenever the filter succeeds. It is target hidden when every operator above depends only on x; the goals $\mathcal{G}$ are used only by a final task-specific check. Complementary composition occurs when $O_x(y_{1:n})$ passes that final check although no individual $O_x(y_i)$ does.*

Loop-invariant generation instantiates the atoms with invariant clauses, composition with conjunction, validity with joint inductiveness, and the filter with bounded Houdini. The transfer property for any other instantiation satisfying Definition 2 is proved in Appendix B.

Definition 2 identifies the conditions needed to reuse CRAFT: responses must decompose into reusable atoms, atoms must admit pooling, a sound domain-specific filter must validate the pooled result, and provenance must map accepted atoms back to responses. Under these conditions, the same inference and training construction applies after replacing the invariant clauses, conjunction, and Houdini checks by the domain's atoms, composition operator, and verifier.

For *specification generation*, atoms can be precondition, postcondition, frame, or invariant clauses; composition is conjunction; and the filter checks syntax, consistency, and the corresponding proof obligations. Separate responses can contribute complementary clauses to one verified contract. For *test-case generation*, atoms are executable tests; composition is set union; the filter removes invalid or duplicate tests; and structural or behavioral coverage supplies a target-independent signal. Separate responses can cover disjoint behaviors even when no one response forms an adequate suite.

These domains share the formal interface and algorithms. Each domain supplies its verifier and measurable training signal.

B Theoretical Properties and Proofs

This section states and proves the properties used by the complete algorithms in Appendix A. Write $\mathcal{L} = (\mathsf{Init}, B, S)$.

B.1 Conjunction of Valid Invariants

Proposition 3 (Closure under conjunction)**.** *If $I_1, \ldots, I_m$ are valid invariants of the same loop, then $\bigwedge_j I_j$ is valid and is at least as strong as every I_j.*

Proof of Proposition 3. For each j, validity gives $\mathsf{Init} \Rightarrow I_j$; hence $\mathsf{Init} \Rightarrow \bigwedge_j I_j = I$. Suppose $I \wedge B$ holds before one execution of S. Then $I_j \wedge B$ holds for every j, so preservation of I_j establishes every I_j afterward. Therefore I is preserved and is a valid invariant.

Because a conjunction implies each conjunct, I is at least as strong as every I_j. □

B.2 Mutual Support Among Arbitrarily Many Clauses

Proposition 4 (Mutual support)**.** *For a finite clause set C, $I(C)$ is valid if*

$$\mathsf{Init} \Rightarrow I(C), \qquad \forall c \in C :\ \mathrm{Pres}_{\mathcal{L},C}(c). \tag{39}$$

No clause must be preserved in isolation.

Proof of Proposition 4. The first premise is exactly establishment of $I(C)$. For preservation, assume $I(C) \wedge B$ before executing S. The second premise establishes each c_j afterward. Since every conjunct then holds in the same successor state, their conjunction $I(C)$ also holds. Thus $\{I(C) \wedge B\} S \{I(C)\}$, and $I(C)$ is valid.

No standalone preservation premise appears in this argument. A clause may use all other conjuncts in the pre-state, so clauses that fail preservation separately may still be preserved jointly. Establishment is different: $\mathsf{Init} \Rightarrow I(C)$ implies $\mathsf{Init} \Rightarrow c_j$ for every j. Hence a candidate false at some loop-entry state cannot be rescued by conjunction. □

B.3 Greatest Jointly Inductive Candidate Subset

Let $C_0 = C$, and let $C_{j+1} \subseteq C_j$ be the set remaining after one ideal Houdini deletion round in Equation (20). Because C is finite, this descending sequence reaches the fixed point $\mathsf{H}_{\mathcal{L}}(C)$.

Theorem 5 (Candidate-relative optimality)**.** *With exact logical decisions, the Houdini fixed point $\mathsf{H}_{\mathcal{L}}(C)$ is jointly inductive and contains every jointly inductive subset of C. Its conjunction is therefore the strongest invariant obtainable by selecting clauses from this pool.*

This is a candidate-relative property of ideal Houdini, not a completeness guarantee for generation or bounded proving. In particular, bounded filtering may discard a valid clause after timeout. Joint checking permits mutual support in preservation, but cannot rescue a clause false at loop entry.

Proof of Theorem 5. At the fixed point, every $c \in \mathsf{H}_{\mathcal{L}}(C)$ satisfies establishment and preservation under the full conjunction $I(\mathsf{H}_{\mathcal{L}}(C))$. Applying Proposition 4 shows that this conjunction is valid.

It remains to prove greatestness. Let $C' \subseteq C$ be any jointly inductive subset. We show by induction over deletion rounds that $C' \subseteq C_j$. The base case $C' \subseteq C_0 = C$ is immediate. Assume $C' \subseteq C_j$, and take any $c \in C'$. Joint establishment of C' gives $\mathsf{Init} \Rightarrow c$, so c cannot be deleted for establishment. Joint preservation gives

$$\{I(C') \wedge B\}\, S\, \{c\}.$$

Since $C' \subseteq C_j$, the antecedent $I(C_j) \wedge B$ implies $I(C') \wedge B$. Strengthening a valid Hoare precondition preserves validity, so

$$\{I(C_j) \wedge B\}\, S\, \{c\}$$

also holds. Therefore no member of C' is deleted in round j, proving $C' \subseteq C_{j+1}$. At termination, $C' \subseteq \mathsf{H}_{\mathcal{L}}(C)$. Because conjunction reverses set inclusion, $I(\mathsf{H}_{\mathcal{L}}(C))$ implies $I(C')$. □

Pooling before filtering. Each separately filtered set $\mathsf{H}_{\mathcal{L}}(A_i)$ is jointly inductive. By closure under conjunction, their union is a jointly inductive subset of $\bigcup_i A_i$. Theorem 5 therefore gives Equation (3). The inclusion can be strict when a clause requires a supporting hypothesis from another response. This ideal property does not ensure empirical monotonicity under finite budgets: pooling changes proof obligations and may trigger timeouts (Appendix M).

B.4 Soundness of Resource-Bounded Houdini

Proposition 6 (Bounded-filter soundness)**.** *Assume* $\mathrm{LoopVerify}_{\mathcal{L},\delta}$ *is sound. If* $\mathrm{Filter}_{\mathcal{L}}^{\delta,h_{\max}}(C)$ *returns a proved fixed point D, then $D \subseteq C$ and $I(D)$ is inductive. Timeout, unknown, and exhaustion of the round budget cannot cause an unproved clause set to be returned.*

Proof. Every deletion round in Equation (21) selects a subset of its input, so $D \subseteq C$. At a returned fixed point, every clause maps to valid in that same family. Verifier soundness in Equation (17) therefore gives $\mathrm{Ind}_{\mathcal{L}}(I(D))$. A timeout or unknown removes the affected clause, and reaching $h_{\max}$ returns the empty result rather than an intermediate set by Equation (23). □

B.5 Supporting Clauses and Credit

Proposition 7 (Coverage credit omits pure support)**.** *Suppose response i contributes a retained clause c that supports another retained clause in the sense of Appendix A, but* $\mathrm{rej}_{\mathcal{N}}(V_i) = \varnothing$*. Then the coverage term assigned to response i is zero, irrespective of whether removing c would prevent the other clause from surviving.*

Proof. The premise gives $R_i = \varnothing$. Therefore $|R_i| = 0$, so the coverage term in Equation (31) is zero. □

This is a formal edge case rather than a material effect in our evaluation. Post-filtering permits supporting clauses from different responses to survive jointly, whereas pre-filtering removes clauses that cannot survive within their own response. Under the longer verifier budget, their compose@k difference ranges from -0.2 to $+0.6$ percentage points across the reported budgets (Table 22). Thus, although the post-filter coverage score does not credit pure indirect support, such cross-response dependencies have negligible aggregate effect on verification in this benchmark.

B.6 Transfer to Verifiably Composable Problems

Proposition 8 (Composable transfer)**.** *For any instantiation satisfying Definition 2, the pooled output $O_x(y_{1:n})$ contains only atoms drawn from the responses and satisfies* Valid_x *whenever filtering succeeds. Moreover, its construction is independent of the hidden goals $\mathcal{G}$.*

Proof. Decomposition and union draw atoms only from $y_{1:n}$; Unique_x removes candidates and the conservative filter returns a subset, establishing provenance to the responses. Filter soundness gives $\mathrm{Valid}_x(O_x(y_{1:n}))$. Finally, every operator in Equation (38) is a function of x and the responses, not $\mathcal{G}$, so hidden goals enter only the final check. □

C Scope and Motivating Example

We focus on numerical loops because their arithmetic equalities and inequalities, including polynomial relations, admit scalable automatic checking. The study considers one braced while loop over scalar C integers, including nonlinear updates. Even this setting can hide nonlinear relations across coupled recurrences. For example:

```c
void f(int k, int z) {
  int x = 1, y = 1, c = 1;
  while (c < k) { c++; x = x*z + 1; y = y*z; }
  /*@ assert 1+x*z-x-z*y == 0; */
}
```

Case study: target-hidden invariant discovery. If the assertion is visible, it can be copied as a loop-invariant candidate. It holds initially because $x = y = 1$. Writing $I(x, y) = 1 + xz - x - zy$, one loop iteration gives

$$I(xz + 1, yz) = 1 + (xz + 1)z - (xz + 1) - z(yz) = zI(x, y),$$

so $I = 0$ is inductive and discharges the exit assertion. Under target hiding, the model sees the initialization and recurrences but not the assertion, and must recover the relation. Archived rollout pools recover three algebraically equivalent forms:

Model	First verified prefix	Rediscovered invariant clause
GPT-5	compose@1	$x - 1 = z(x - y)$
DeepSeek V4-Flash	compose@4	$x(z - 1) = zy - 1$
Claude Sonnet-4.6	compose@8	$x(z - 1) = yz - 1$

All three clauses rearrange to the hidden assertion. None of the ten raw responses in these pools passes as a whole on this program; pooled filtering removes incompatible siblings and retains the useful relation.

Case study: composition without cross-response support. To separate target-proof composition from both sampling a better standalone response and mutual support during filtering, we replay the first four archived GPT-5-nano responses for the following task. We apply the target-hidden $\mathrm{Filter}_{\mathcal{L}}$ used in evaluation first to each response separately and then, following the pooled order of Appendix M, once to their merged clause union. The restored-target judge is applied only after filtering. The assertion is shown for clarity but is hidden from every response.

```
/*@ requires n >= 0; */
void f(int n) {
  int x = n, y = 0;
  while (x > 0) {
    y = y + 1;
    x = x - 1;
  }
  /*@ assert y == n; */
}
```

Table 2 reports both the relevant admitted proposal from each response and the result of filtering that response alone.

Response	Relevant proposal	Independent survivors	Target
1	$x \geq 0,\ y \geq 0$	$x \geq 0,\ y \geq 0$	fail
2	—	$\varnothing$	fail
3	$x = n - y,\ y = n - x$	$x = n - y,\ y = n - x$	fail
4	$x \geq 0,\ y \geq 0$	$x \geq 0,\ y \geq 0$	fail
Pooled 1–4	$x \geq 0,\ y \geq 0,\ x = n - y,\ y = n - x$		verified

Table 2: A cross-response composition case without mutual support: every pooled survivor already survives independent filtering, yet no response alone verifies the target.

Each of the four distinct pooled clauses also survives when filtered alone, so pooling does not rescue any clause by supplying a missing inductive hypothesis. The complete pooled annotation is

```
/*@
  loop invariant x >= 0;
  loop invariant y >= 0;
  loop invariant x == n - y;
  loop invariant y == n - x;
*/
```

The proof-minimal cross-response core is $x \geq 0$ from Response 1 (also duplicated by Response 4) and $y = n - x$ from Response 3. Each clause alone fails the restored-target judge. Together, the exit condition gives $x \leq 0$, the bound gives $x = 0$, and the relation yields $y = n$. Thus the gain is genuine composition for proving the target, not a better standalone response or cross-response support during inductiveness filtering.

Case study: composition through cross-response support. The preceding case needs composition only to prove the target. A second archived GPT-5-nano replay shows the stronger mechanism from Proposition 4: pooling can also make a clause provably inductive. As before, the assertion is restored only after target-hidden filtering.

```
extern int unknown(void);
void f(int n) {
  int x = 1, m = 1;
  while (x < n) {
    if (unknown()) m = x;
    x = x + 1;
  }
  if (n > 1) { /*@ assert m >= 1; */ }
}
```

Replaying its first four responses gives:

Response	Relevant proposal	Independent survivors	Target
1	$m \leq x$	$m \leq x$	fail
2	$x \geq 1$	$x \geq 1$	fail
3	—	$\varnothing$	fail
4	$m \geq 1$	$\varnothing$	fail
Pooled 1–4	$m \leq x,\ x \geq 1,\ m \geq 1$		verified

Response 4's $m \geq 1$ cannot be established inductive alone: after the nondeterministic assignment $m = x$, preservation requires $x \geq 1$. Response 2 supplies that hypothesis, so pooled filtering retains both clauses. The complete composed annotation is

```
/*@
  loop invariant m <= x;
  loop invariant x >= 1;
  loop invariant m >= 1;
*/
```

All four independently filtered responses fail, whereas the pooled annotation verifies. This is cross-response support during filtering, not merely complementary facts combined at the final target check.

D Implementation Details

Table 3 lists the implemented defaults. SFT synthesis, RL, and inference share Algorithm 1. RL filters the union once, then uses clause provenance to assign survivors back to their proposing rollouts before computing credit (Appendix M).

Symbol	Value	Role
r_{exec}	12	requested input executions; doubled for oracle-driven bodies
$\mathcal{X}_{\text{tier}}$	20; $[-64, 64]$	fixed near-input schedule and clamping interval
q_{far}	3, plus 2 ordered multi-parameter probes	deterministic far probes with magnitude at most 512
Δ_{rel}	dense: $\pm\{1, 2\}$; terminal: $\pm\{1, 2, 3, 5, 8\}$	local relational perturbations
h_{dense}	3	successor window used to establish a relation base
b_{rel}	96	cap on relation bases stratified across positives
h_{exit}	24	continuation horizon after a witnessed exit
$b^-_{\text{rel}}, b^-_{\text{exit}}$	48, 12	relational and post-exit negative-trace budgets
$u_{\max}$	2×10^6	execution-iteration safety cap
w_c	1.0	coverage weight
$m_{\max}$	20 clauses	cap applied independently to each response
$\mathcal{K}$	$\{1, 4, 8, 16, 32\}$	reported local-model response budgets
n_{probe}	128; 100; 32	saved responses per task (8B; A3B/Llama; remaining base models)
ϑ	1.0	decoding temperature except for Loopy (Table 21)
p	1.0	nucleus-sampling probability
L_{api}	8,192 tokens	maximum API-model completion length
L_{local}	2,048 tokens	maximum local-model completion length
g_{sft}	16 responses	group sampled per SFT strengthening round
t_{sft}	5 rounds	maximum per loop; at most 80 responses
τ	0.5	SFT acceptance threshold on composed negative coverage
$h_{\max}$	500	maximum Houdini verification rounds
$\mathcal{P}_{\text{verify}}$	Alt-Ergo and Z3	prover portfolio; Typed memory model
δ	5 seconds per goal	per-prover goal budget used by LoopVerify
$\delta_{\text{short}}, \delta_{\text{long}}$	300, 900 seconds	verification caps in the pre-/post-filtering comparison

Table 3: Parameter values for execution sampling (§4.2), SFT synthesis (§4.3), reward computation (§4.4), inference, and verification and evaluation.

The fixed near-input tier order is $\langle 0, 1, 2, -1, 3, 5, 8, -2, 10, 17, 4, -3, 25, 6, 40, 7, -5, 13, 50, 9\rangle$.

D.1 Loop-Verifier Realization

The implementation generates establishment and preservation goals for each clause in the ordered current family $D = \langle d_1, \ldots, d_m\rangle$. Establishment of d_k may use earlier annotations d_j, $j < k$, as hypotheses. Preservation uses the full current family as loop-head hypotheses and may use earlier annotations in the successor state. These hypotheses are available during the current round regardless of the status of their own goals. The verifier simplifies and splits the obligations by control-flow path. A clause receives valid only when every generated establishment and preservation subgoal is proved; timeout records an exhausted prover budget, and all other outcomes map to unknown. Houdini reruns the survivors after each deletion round.

D.2 NEGATIVE-TRACE SAMPLING

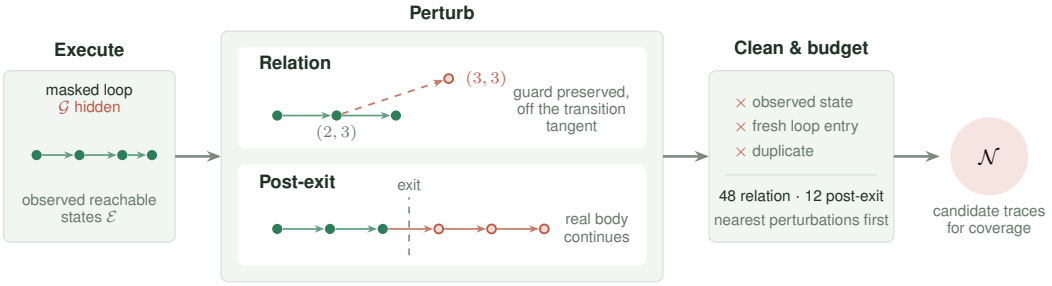

Figure 6: **Construction of target-hidden candidate negative traces.** Concrete executions provide observed reachable states. Two transformations probe relational structure and behavior past a genuine exit. Conservative cleaning and per-family budgets select these two structured families; random perturbations fill unused capacity (not shown). The resulting $\mathcal{N}$ supplies coverage for SFT acceptance and RL rewards without exposing $\mathcal{G}$; it does not certify unreachability.

Algorithm 5 specifies the structured sampler's selection order. The routines below define its execution, proposal, cleaning, and budget rules. These are candidate negatives, not certified unreachable states. Neither generation nor cleaning consults hidden targets.

Algorithm 5 Target-hidden candidate-negative sampling

Input: masked loop $\mathcal{L}$, input schedule U, configuration Θ
Output: observed states $\mathcal{E}$, candidate traces $\mathcal{N}$

$\widetilde{\mathcal{L}} \leftarrow \mathrm{Determinize}(\mathcal{L})$ ▷ *execution only*
$(T, O, a) \leftarrow \mathrm{Trace}(\widetilde{\mathcal{L}}, U, \Theta)$ ▷ *a: any run capped*
$\mathcal{E} \leftarrow \mathrm{UniqueHeads}(T)$; $Z \leftarrow \varnothing$; $\mathcal{N} \leftarrow \langle\rangle$
$M \leftarrow \mathrm{Movable}(\mathcal{L})$; $Q \leftarrow \mathrm{StratifiedBases}(\mathcal{E}, 96)$
$b \leftarrow \mathrm{PerturbationBlocked}(\mathcal{L})$
if $\neg b$ **then**
  $C \leftarrow \mathrm{RelationProposals}(Q, T, M, a)$
  $C \leftarrow \mathrm{Clean}(C, \mathcal{E}, Z)$
  $R \leftarrow \mathrm{RoundRobinNearest}(C, 48)$
  append $\langle s \rangle$ for each $s \in R$ to $\mathcal{N}$; add their keys to Z
$e \leftarrow 0$
for each run u in increasing order with $e < 12$ **do**
  $\nu \leftarrow \mathrm{Clean}(O_u, \mathcal{E}, Z)$
  if $\nu \neq \varnothing$ **then** append ν to $\mathcal{N}$, add its keys to Z, and increment e
if $\neg b$ **then** $\mathcal{N} \leftarrow \mathcal{N} \| \mathrm{RandomFill}(Q, M, 60 - |\mathcal{N}|, \mathcal{E}, Z)$
return $(\mathcal{E}, \mathcal{N})$

Execution and state identity. Determinize replaces each supported zero-argument oracle call site with a fresh sampled integer parameter, fixed throughout a run. This explores a restricted set of oracle behaviors; the original program is still used for generation and proof. A state key includes its current valuation and both function-entry and loop-entry snapshots. Perturbations preserve these snapshots, so states from different contexts are not conflated. The requested run count is 12, doubled for oracle-driven bodies; the input schedule and execution cap are in Table 3. T records entry, sampled heads, and exits with run/iteration coordinates. After an uncapped exit, the instrumented executor replays the body for at most 24 steps without the original guard to form O_u. Bodies containing `return`, `goto`, or labels disable this replay. Capped runs contribute no post-exit states; detected abnormal runs are skipped. If no observed states remain, sampling reports failure rather than inventing traces.

Perturbation eligibility and bases. M contains tracked variables assigned by the body. If none are found, the implementation falls back to tracked non-parameter variables, or all tracked variables if

that set is also empty. Untracked persistent assignments, unsupported body calls, or unsupported local pointer/array declarations block both relational and random perturbations, but do not by themselves discard executed post-exit traces. Ordinary block-local scalar temporaries are not treated as persistent state. Q allocates at most 96 bases round-robin across runs and entry contexts, keeping dense prefixes, endpoints, and evenly spaced positions where quotas permit.

Relation proposals. A base must have the next three head positions recorded, or be an observed terminal with a guard-true predecessor and guard-false current state. Terminal bases are added even if omitted from Q, but are disabled when any run was capped. For each base, perturb one axis by d, or a pair by (d, d) or $(d, -d)$. Interior offsets are $d \in \pm\{1, 2\}$; terminal offsets are $d \in \pm\{1, 2, 3, 5, 8\}$. Retain a proposal only if every coordinate stays within its context's sampled minimum/maximum envelope and the guard has the same known truth value. Reject changes that are an integer multiple of the observed forward step (or the predecessor step if no successor is recorded). This tangent test avoids obvious movements along witnessed dynamics; it is not a reachability decision. Envelope checks prevent simple sampled marginal bounds from separating this family, but do not constrain post-exit or random candidates.

Cleaning and its limits. Clean removes keys in $\mathcal{E}$, previously selected keys Z, and duplicates within its input. It also drops a state if the entry screen succeeds: parameter values agree with their function-entry values, every retained local-initialization equality evaluates to true, and the precondition does not evaluate to false. An unknown initialization equality makes this screen fail; an unknown precondition does not. Consequently the screen is not an exact initialization-reachability test. For a post-exit group, cleaning removes individual states while preserving the order of the remainder; an empty group is discarded.

Exclusion from $\mathcal{E}$, failure of the entry screen, and departure from a local tangent do not prove unreachability. Likewise, overriding an exit creates a counterfactual execution, but its states may occur on another legal execution. No general unreachability solver is called. Under sound inductive verification and consistent state-evaluation semantics, a truly reachable state cannot be rejected by a valid invariant. Such contamination can nevertheless dilute coverage and affect relative rewards or SFT acceptance. The final proof guarantee is supplied by the verifier, not by the sampler's labels.

Selection, random fill, and scoring units. Relation candidates are bucketed by interior/terminal status, changed axes, and change signs. Stable round-robin selection takes up to 48 candidates, preferring smaller ℓ_1 changes within each bucket. Only selected keys are committed to Z. Next, up to 12 nonempty post-exit groups are selected in run order. Thus cross-family deduplication gives priority to relation, then post-exit, then random fill.

$\mathrm{RandomFill}$ draws a base uniformly from Q, chooses one or two axes uniformly (one if only one is available), and samples a nonzero integer offset independently per axis. It first uses magnitudes 1–34, then 35–128 if needed, with seed $seed \operatorname{xor} \texttt{0x4C4F4F50}$. Each stage permits at most $\max(1024, 100q)$ attempts for the requested remaining capacity q. Candidates use the same observed-state, entry, and duplicate checks, but not the relation-specific envelope, guard, or tangent constraints. Accepted states immediately enter Z and each forms a singleton trace. The random baseline uses this routine alone with $q = 60$. Neither mode is guaranteed to fill all 60 slots when admissible candidates are scarce.

Every selected singleton or cleaned continuation contributes one coverage unit, rejected if any retained clause is definitely false at any witness. This avoids counting a long continuation repeatedly, although longer traces still offer more rejection opportunities. Unknown evaluations do not reject a trace. Empty-negative handling is specified in Algorithm 4 and Section 4.3.

D.3 Clause Normalization and Optimization Settings

Appendix A.7 defines the group-normalized advantage, token-level importance ratio, sampled conditional KL term, and clipped GRPO objective. Clause deduplication, which forms the clause set scored above, normalizes comparison direction, equality sides, parentheses, immediate commutative operands, harmless identities, and Boolean association, but avoids transformations that require additional arithmetic assumptions. Repeated clauses therefore collapse without affecting coverage, and a unique zero-coverage clause costs nothing because it may support another clause or the hidden target.

Coverage lies in $[0, 1]$; clauses beyond the admission cap incur no reward penalty. Empty-negative groups use the binary fallback in Algorithm 4.

E GENERATION PROMPT

The canonical evaluation template appears below; every target is stripped before insertion, and general annotation rules are supplied by the common system prompt. Both the SFT and RL data use this single prompt pair.

Generation prompt

```
You are given a C function. Produce the strongest inductive ACSL loop
invariants that capture the loop's behavior (conservation/relational laws
+ tight progress bounds). Output ONLY invariant lines, each formatted
exactly as:
loop invariant <expr>;
Output at most 20 invariant lines.

Program:
{program}
```

The accompanying system prompt appears verbatim below.

System prompt

```
You are a formal verification expert specializing in ACSL loop invariant
synthesis for Frama-C/WP.

## LOOP INVARIANT DEFINITION

A loop invariant is a property that:
1. Holds before the first loop iteration (initialization).
2. Is preserved by every loop iteration when the loop guard is true
   (inductiveness).
3. Combined with loop exit (`!guard`), helps establish required
   loop-exit facts.

## RULES

- Add one core conservation/relational equality that explains loop
  updates (state transition law).
- Add minimal progress bounds for loop counters and key arithmetic
  variables.
- Prefer strong invariants; avoid weak/tautological ones.
- Do not add unrelated invariants just to increase count.
- MUST NOT modify or rewrite any `unknown()`/`unknownN()` call.
- `\at(v, Pre)` can ONLY be used with function parameters, never with
  local variables initialized inside the function.
- `\at(v, LoopEntry)` denotes the value of v at the start of the loop
  and can ONLY be used with local variables initialized before the
  loop (using v = unknown();).
- Loop guard is NOT an invariant; weaken strict guards to non-strict
  bounds at exit.
- Parenthesize mixed equality and relational expressions so their
  grouping is explicit.
- Use `loop invariant` only as a line prefix, never inside expressions.
- Emit scalar invariant expressions only; do not declare predicates,
  logic functions, axioms, or lemmas.
- Do NOT use the ternary `? :` operator; split cases with `==>` instead.
- `^` is bitwise XOR in C/ACSL, not exponentiation --- write `x*x*x`
  for x cubed.
- Do NOT place a C boolean directly in arithmetic (`x + (a > b)` is
  invalid); use `==>` to split cases.
- Only use variables declared in the given function.
- Operators: comparison `==` `!=` `<` `<=` `>` `>=`; logical `&&` `||`
  `!` `==>` `<==>`; arithmetic `+` `-` `*` `/` `%`.

## ACSL INVARIANT SYNTAX

Allowed expression forms (all verified by Frama-C/WP):

1. **Bounds**: `0 <= i`, `i <= n`, `x >= 1`
2. **Linear equality**: `i + c == n`, `2 * s == i * (i - 1)`,
   `x == q * y + r`
3. **Polynomial equality**: `x == n * n * n`,
   `6 * s == i * (i + 1) * (2 * i + 1)`
4. **Relational identity** (multi-var algebraic law):
   `1 + x * z - x - z * y == 0`, `p * s - r * q == 1`
5. **Implication**: `i > 0 ==> s >= i`, `a != 0 ==> q >= 1`
6. **Modular**: `i % 5 == 0`, `r >= 0 && r < y`
7. **Bitshift** (power-of-2 only): `p == 1 << i`, `s == (1 << i) - 1`
8. **Conjunction**: `c >= 1 && c <= k`
9. **\at for params**: `n == \at(n, Pre)` (function parameters only)

Forbidden:
- `^` for exponentiation (it is XOR) --- use repeated `*` or `<<` for
  powers of 2.
- `\product`, `\sum`, `\factorial` --- not ACSL scalar operators.
- `? :` ternary --- rewrite with `==>`.
- `\old(x)` --- use `\at(x, Pre)`.
- Type casts `(int)`, `(long)` --- unnecessary in ACSL logic.
- Function calls, predicates, lemmas, axioms --- emit only scalar
  expressions.

Operators (complete list):
  comparison: `==` `!=` `<` `<=` `>` `>=`
  logical:    `&&` `||` `!` `==>` `<==>`
  arithmetic: `+` `-` `*` `/` `%` `<<` `>>`

## OUTPUT

Return ONLY invariant lines, one per line, formatted exactly as:
`loop invariant <property>;`
Do not return code fences, `loop assigns`, the surrounding C function,
or explanations.
```

F TRAINING INTERFACE

The training service is packaged as a self-contained, CPU-only container with gcc, the verifier, Why3, and its prover backend. It accepts program source, an array of model responses, one of the

three reward variants defined below, and either random or structured negative sampling. The reward and sampler choices are independent. The response reports each rollout's reward together with its coverage. Responses may be raw LLM text, invariant lists, or annotated code.

G Training–Evaluation Overlap Audit

We compare the historical RL (4,992 records and 4,992 distinct program sources) and SFT (33,346 records and 31,865 distinct program sources) training sets used in the reported experiments against all 832 evaluation programs after target removal. This audit does not describe the subsequently released 9,024-record RL set (Appendix H). Under three increasingly aggressive normalizations—exact text, alpha-renamed identifiers, and alpha-renaming with integer constants abstracted—no training program matches an evaluation program (Table 4).

We separately measure coarse structural overlap. A structural cell combines the control-flow profile, guard kind, nonlinearity, nondeterminism, and a variable-count band. The evaluation set occupies 122 such cells. RL and SFT share 88 and 41 of them, respectively; their union shares 92 cells. Equivalently, 702 (84.4%), 529 (63.6%), and 754 (90.6%) of the 832 evaluation programs fall in a cell represented by RL, SFT, and their union. The record-weighted total-variation distances over cells are 0.422 for RL, 0.651 for SFT, and 0.596 for their union, so the training and evaluation distributions are related but not identical. Table 5 gives interpretable feature marginals behind this joint-cell comparison.

	RL	SFT	RL ∪ SFT
Records	4,992	33,346	38,338
Distinct program sources	4,992	31,865	35,170
Exact-text matches	0	0	0
Alpha-renamed matches	0	0	0
Alpha-renamed, constant-abstracted matches	0	0	0
Shared evaluation cells	88 / 122	41 / 122	92 / 122
Evaluation programs in shared cells	702 (84.4%)	529 (63.6%)	754 (90.6%)
Training records in evaluation-supported cells	4,992 (100%)	27,275 (81.8%)	32,267 (84.2%)
TV distance over structural cells	0.422	0.651	0.596

Table 4: Instance and structural-distribution audit of the historical training sets against the 832 target-removed evaluation programs. Distribution statistics count training records, matching their sampling frequency.

Feature (% of records/programs)	Evaluation	RL	SFT
Constant guard	24.6	7.5	4.6
Single-variable guard	31.4	31.6	44.0
Variable-vs.-variable guard	37.3	57.3	50.6
Compound guard	6.7	3.7	0.8
Nonlinear	2.5	2.5	10.8
Nondeterministic	28.6	30.3	14.9
≤ 2 variables	35.1	21.2	46.2
3–4 variables	33.9	31.7	12.0
≥ 5 variables	31.0	47.1	41.8

Table 5: Marginal structural distributions of evaluation programs and historical training records. The structural-cell TV distances in Table 4 use the joint features rather than these marginals independently.

H Training-Data Curation

The experimental RL and SFT training sets use the target-hidden scalar-integer single-loop interface. SFT targets are inductive invariant sets synthesized by the multi-round procedure in Section 4.3 and accepted by the loop verifier. The only learned generator is the open-weight model being trained;

target acceptance and RL reward computation use program execution and verification rather than labels or critiques from an external model. Table 6 distinguishes the experimental sets from the current RL release. The latter contains 9,024 unchanged records selected from the raw pool, but was not used to produce the reported historical RL results. The overlap and structural statistics above refer only to the experimental sets and must not be transferred to this newer release.

Training set	Size and composition
Experimental RL	4,992 records over 1,131 distinct loop shapes
Experimental SFT	33,346 records over 31,865 distinct program sources
Experimental RL–SFT overlap	1,687 distinct program sources
Current RL release	9,024 records (64×141); not used for reported results

Table 6: Training data used in the experiments and the subsequent RL release.

I Experimental Protocol Details

The evaluation workload composition is given in Section 5.1; the 466 Loopy programs are the integer subset of its original 469-program corpus (three floating-point programs excluded), and unsupported inputs remain failures in the common denominator.

Before model invocation, we remove assert and ensures predicates, executable assertion calls, and conventional error-label names; preconditions and executable semantics remain visible. CRAFT evaluation runs use the canonical generation template, extractor, canonical-key deduplication, and Houdini implementation; external tools retain their own orchestration. Each model retains its serving interface and chat template. All API and locally served CRAFT checkpoints use reasoning-disabled decoding. Decoding uses temperature ϑ, nucleus parameter p, and completion caps L_{api} and L_{local} for API and local models, respectively, without an additional top-k, repetition penalty, re-roll, or target feedback. Each RQ1 curve reuses the same saved response pool per task. The model-specific pool sizes and the common reported budget grid are listed in Table 3; every pool contains at least $k_{\max}$ responses. For each task t, both metrics use the same prefix $y_{t,1:k}$ of its saved response pool, in the original saved order. Let $s_{t,i} \in \{0,1\}$ indicate whether response i succeeds alone under the final verifier, and let $c_{t,k} \in \{0,1\}$ indicate whether filtering and composing the clauses from $y_{t,1:k}$ succeeds under that verifier. For an evaluation set of N tasks, we report

$$\text{pass}\,@k = \frac{100}{N} \sum_{t=1}^{N} \max_{1 \le i \le k} s_{t,i},$$

$$\text{compose}\,@k = \frac{100}{N} \sum_{t=1}^{N} c_{t,k}.$$

These are percentages of tasks solved using the first k responses. We do not average over alternative subsets of the saved pool or use the combinatorial pass@k estimator. All budgets reuse the same saved order, so the evaluated prefixes are nested. Appendix K discusses the resulting sampling variance.

Training configuration. Tables 7 and 8 record the supplied SFT and RL run configurations. All reported SFT checkpoints are trained for three epochs, and all reported RL checkpoints are trained for five epochs. Reward and sampler constants are reported separately in Table 3.

Field	Value
Learning rate η_{sft}	1×10^{-5}
Epochs e_{sft}	3
Batch (per device $\times$ accumulation $\times$ GPUs)	global batch 32; $2 \times 2 \times 8$ or $1 \times 4 \times 8$
Sequence cutoff L_{sft}	2,048
DeepSpeed	ZeRO-3
Scheduler	cosine

Table 7: SFT training configurations.

Field	Value
Batch size b_{rl}	64
Learning rate η_{rl}	1×10^{-6}
KL coefficient β	0.001
Advantage stabilizer δ_{adv}	10^{-6}
Rollouts per prompt g_{rl}	8
Tensor parallelism p_{tp}	1–2
GPU memory utilization μ_{gpu}	0.3–0.5
Epochs e_{rl}	5

Table 8: RL training configurations.

J Complete Experimental Data

This section collects the complete available measurements behind the main aggregates: full local-model probe curves, SFT and RL checkpoints, RL directly from Bare, reasoning-disabled API models, and specialized tools. We report every archived sampling budget for each experiment and retain benchmark- stratum results whenever they are available. A guide: §J.1–J.2 give the Bare and SFT probe curves behind RQ1; §J.3 gives the RL checkpoints behind RQ2 and RQ4; §J.5–J.6 give the clause-cap and reward ablations behind RQ4 and RQ5; §J.7 evaluates the negative-coverage surrogate; §J.8–J.9 give the cross-model and tool comparisons behind RQ3.

J.1 Probe Curves

In addition to the four backbones plotted in Figure 2, we evaluate Qwen3-1.7B and Qwen3-30B-A3B. Table 9 retains all six checkpoints and their per-stratum results. Composition is not monotonic in total parameter count: Qwen3-30B-A3B trails the dense 8B and 14B checkpoints at compose@32.

Table 9: Complete Bare probe results by benchmark stratum and sampling budget. Each cell reports a percentage; All / 832 is the whole-workload result. Bold marks the best value in each column at a fixed budget.

Model	k	Linear / 316 pass	Linear / 316 comp	NLA / 50 pass	NLA / 50 comp	Loopy / 466 pass	Loopy / 466 comp	All / 832 pass	All / 832 comp
Qwen3-1.7B	1	5.50	19.30	2.22	6.00	6.32	21.03	5.76	19.47
	4	9.70	29.75	4.77	8.00	12.32	30.69	10.87	28.97
	8	12.24	33.23	**5.97**	8.00	15.54	35.62	13.71	33.05
	16	15.14	35.76	**7.05**	10.00	18.84	37.77	16.72	35.34
	32	18.30	38.29	**7.79**	**14.00**	22.13	40.13	19.81	37.86
Qwen3-4B	1	5.04	33.54	0.46	**8.00**	5.95	35.19	5.27	32.93
	4	10.36	44.94	1.59	10.00	10.53	47.21	9.93	44.11
	8	13.14	50.00	2.66	10.00	13.12	50.00	12.50	47.60
	16	15.73	52.53	3.93	12.00	15.95	53.65	15.14	50.72
	32	18.31	54.11	5.15	12.00	18.96	56.22	17.88	52.76
Qwen3-8B	1	**6.13**	37.97	**2.25**	6.00	7.07	41.20	6.43	37.86
	4	**13.06**	50.95	**4.88**	6.00	14.25	55.15	13.24	50.60
	8	**17.40**	54.75	5.77	10.00	18.52	58.58	17.33	54.21
	16	**21.84**	56.65	6.00	10.00	23.14	61.16	21.62	56.37
	32	**25.63**	58.23	6.00	10.00	28.33	62.66	25.96	57.81
Qwen3-14B	1	6.06	**48.10**	1.08	4.00	**9.47**	**44.64**	**7.67**	**43.51**
	4	11.66	**61.39**	2.85	10.00	**17.09**	**61.37**	**14.17**	**58.29**
	8	15.01	**64.56**	3.79	**12.00**	**21.49**	**64.81**	**17.97**	**61.54**
	16	18.82	**66.14**	4.50	**14.00**	**26.11**	**67.38**	**22.04**	**63.70**
	32	22.91	**66.46**	5.08	**14.00**	**30.57**	**68.67**	**26.13**	**64.54**
Qwen3-30B-A3B	1	1.33	16.14	0.06	0.00	1.59	20.17	1.40	17.43
	4	4.76	29.43	0.23	4.00	4.84	34.33	4.53	30.65
	8	8.33	36.71	0.45	4.00	7.97	39.91	7.65	36.54
	16	13.27	43.35	0.82	8.00	12.60	45.28	12.15	42.31
	32	18.82	43.99	1.38	8.00	18.38	47.64	17.53	43.87
Llama 3.1-8B	1	0.57	28.16	0.02	6.00	0.69	30.04	0.61	27.88
	4	2.15	48.73	0.08	**12.00**	2.39	47.85	2.16	46.03
	8	3.97	54.43	0.16	**12.00**	4.08	53.22	3.81	51.20
	16	6.88	55.06	0.32	12.00	6.54	53.65	6.30	51.68
	32	10.88	55.06	0.64	12.00	9.98	54.51	9.76	52.16

Table 10: Per-GPU decoding throughput of the four backbones used in training experiments, measured with vLLM on A800-80G GPUs using response budget n_{probe}, temperature ϑ, and completion cap L_{local}.

Model	tok/s/GPU
Qwen3-4B	7,445
Qwen3-8B	5,833
Qwen3-14B	3,168
Llama 3.1-8B	5,135

J.2 SFT Probe Curves

The SFT checkpoints use rejection-filtered synthetic targets and are evaluated under the same 832-task protocol as the base probe, using response budget n_{probe}, temperature ϑ, per-obligation prover budget δ, and the current prompt. All four checkpoints are evaluated after three epochs and use the SFT synthesis and verification protocol defined in Section 4.3.

Table 11: Complete SFT-checkpoint probe results by benchmark stratum and sampling budget. Each cell reports a percentage; All / 832 is the whole-workload result. Bold marks the best value in each column at a fixed budget.

		Linear / 316		NLA / 50		Loopy / 466		All / 832	
Model	k	pass	comp	pass	comp	pass	comp	pass	comp
Qwen3-4B + SFT	1	43.69	65.82	2.08	8.00	42.34	66.31	40.43	62.62
	4	58.55	78.48	4.81	10.00	58.38	76.82	55.23	73.44
	8	63.36	81.01	6.23	12.00	63.47	79.83	59.99	76.20
	16	67.55	82.28	7.99	12.00	67.77	80.90	64.09	77.28
	32	71.22	82.28	10.36	12.00	71.54	81.97	67.74	77.88
Qwen3-8B + SFT	1	45.69	68.40	**3.60**	8.00	43.50	67.00	41.93	63.90
	4	60.99	77.20	6.26	10.00	59.41	78.30	56.82	73.80
	8	65.36	80.10	7.34	10.00	64.00	80.70	61.11	76.20
	16	69.01	81.30	8.44	12.00	67.78	81.50	64.68	77.30
	32	72.34	81.60	9.64	12.00	71.28	82.40	67.98	77.90
Qwen3-14B + SFT	1	**50.18**	70.25	3.54	**12.00**	**47.12**	**69.53**	**45.66**	**66.35**
	4	**63.93**	**81.33**	**7.27**	**16.00**	**61.46**	**80.04**	**59.14**	**76.68**
	8	**68.00**	**82.28**	**9.79**	**16.00**	**66.21**	**82.40**	**63.50**	**78.37**
	16	**71.42**	**82.91**	**12.06**	**16.00**	**70.50**	**83.48**	**67.34**	**79.21**
	32	**74.17**	**82.91**	**13.52**	**16.00**	**74.25**	**83.69**	**70.57**	**79.33**
Llama 3.1-8B + SFT	1	43.60	67.09	2.22	8.00	40.26	63.09	39.24	61.30
	4	56.03	75.32	4.51	14.00	55.41	72.96	52.59	70.31
	8	60.05	77.53	6.15	14.00	60.15	76.39	56.86	73.08
	16	63.64	78.48	7.99	14.00	63.96	77.47	60.47	74.04
	32	66.90	78.80	9.57	14.00	67.34	77.90	63.70	74.40

Table 12: Per-GPU decoding throughput of the SFT checkpoints, measured with vLLM on A800-80G GPUs using 100 responses per task, temperature ϑ, and completion cap L_{local}.

Model	tok/s/GPU
Qwen3-4B + SFT	7,445
Qwen3-8B + SFT	5,833
Qwen3-14B + SFT	3,168
Llama 3.1-8B + SFT	5,135

J.3 Reinforcement-Learning Results

Tables 13 and 14 report the available RL results from Bare and SFT initializations, respectively, with benchmark-stratum breakdowns. The matched Qwen3-8B RL-from-Bare curve and Full decompose credit reward controls appear in Table 18.

Table 13: Complete results for three RL checkpoints obtained by applying RL directly to the corresponding Bare models, aligned with Tables 9 and 11 (%). Each cell reports a percentage; All / 832 is the whole-workload result. Bold marks the best value in each column at a fixed budget. The matched Qwen3-8B RL curve appears in Table 18.

		Linear / 316		NLA / 50		Loopy / 466		All / 832	
Model	k	pass	comp	pass	comp	pass	comp	pass	comp
Qwen3-4B + RL	1	2.20	52.20	0.10	10.00	1.60	50.60	1.70	48.80
	4	4.10	63.60	0.20	10.00	3.30	61.60	3.40	59.30
	8	5.30	66.50	0.50	12.00	4.30	65.20	4.50	62.50
	16	6.60	69.30	1.00	12.00	5.60	67.20	5.70	64.70
	32	8.20	70.30	2.00	12.00	7.30	68.70	7.30	65.90
Llama 3.1-8B + RL	1	6.10	51.90	0.00	6.00	3.80	46.40	4.50	46.00
	4	10.80	58.90	0.00	8.00	8.00	55.20	8.60	53.70
	8	12.70	60.40	0.00	8.00	10.00	58.60	10.50	56.20
	16	14.90	62.30	0.00	8.00	11.80	59.70	12.20	57.60
	32	16.80	63.30	0.00	10.00	13.50	61.20	13.90	58.90
Qwen3-14B + RL	1	**9.50**	**66.80**	**0.60**	**14.00**	**12.00**	**70.00**	**10.40**	**65.40**
	4	**14.80**	**72.80**	**1.70**	**14.00**	**18.10**	**74.70**	**15.80**	**70.30**
	8	**17.90**	**74.40**	**2.40**	**14.00**	**20.90**	**77.00**	**18.70**	**72.20**
	16	**21.30**	**75.60**	**3.00**	**14.00**	**23.60**	**78.50**	**21.50**	**73.60**
	32	**24.70**	**76.60**	**4.00**	**14.00**	**26.00**	**79.40**	**24.20**	**74.40**

Table 14: Complete SFT+RL-checkpoint results by benchmark stratum and sampling budget. Each cell reports a percentage; All / 832 is the whole-workload result. Bold marks the best value in each column at a fixed budget.

Model	k	Linear / 316 pass	Linear / 316 comp	NLA / 50 pass	NLA / 50 comp	Loopy / 466 pass	Loopy / 466 comp	All / 832 pass	All / 832 comp
Qwen3-4B + SFT+RL	1	35.3	69.9	2.8	8.0	35.4	69.7	33.4	66.1
	4	50.0	76.9	4.1	12.0	48.1	76.0	46.1	72.5
	8	54.2	79.1	4.5	**14.0**	51.9	77.9	49.9	74.5
	16	58.2	79.1	5.0	**14.0**	55.2	79.0	53.3	75.1
	32	62.3	80.1	6.0	**14.0**	57.9	79.2	56.5	75.6
Qwen3-8B + SFT+RL	1	32.77	75.00	2.78	**14.00**	32.43	71.24	30.78	69.23
	4	46.39	**79.75**	3.83	**14.00**	45.69	78.76	43.44	75.24
	8	51.03	**80.70**	3.99	**14.00**	49.72	80.90	47.47	**76.80**
	16	54.65	**80.70**	4.00	**14.00**	52.92	**82.40**	50.63	**77.64**
	32	57.78	**81.33**	4.00	**14.00**	55.85	**83.26**	53.47	**78.37**
Qwen3-14B + SFT+RL	1	**45.1**	**75.3**	**4.9**	10.0	**41.6**	**74.2**	**40.7**	**70.8**
	4	**57.2**	79.1	**7.3**	12.0	**54.9**	**80.3**	**52.9**	**75.7**
	8	**60.3**	80.1	**8.3**	12.0	**59.6**	**81.5**	**56.8**	**76.8**
	16	**63.4**	**80.7**	**9.0**	**14.0**	**63.4**	82.0	**60.1**	77.3
	32	**66.8**	80.7	**10.0**	**14.0**	**66.3**	82.0	**63.1**	77.4
Llama 3.1-8B + SFT+RL	1	38.16	67.7	1.62	8.0	34.12	67.0	33.70	63.7
	4	50.78	74.7	3.15	**14.0**	47.06	74.0	45.83	70.7
	8	54.35	76.6	3.69	**14.0**	51.38	76.4	49.64	72.7
	16	57.25	76.9	3.96	**14.0**	55.40	76.8	53.01	73.1
	32	60.18	77.2	4.00	**14.0**	59.44	76.8	56.39	73.2

J.4 Effect of Target Visibility at Inference

We compare target-hidden and target-visible inference using the same Qwen3-8B SFT+RL checkpoint, which was trained entirely under the target-hidden protocol. The target-visible condition exposes the target assertions and postconditions $\mathcal{G}$ during generation; all other generation and verification settings and response budgets are held fixed. Neither condition uses iterative verifier-feedback refinement, and the checkpoint receives no additional training for target visibility. The target-visible results were reported as combine@k; we use compose@k here for the same pooled clause-filtering procedure. The target-hidden reference is the Qwen3-8B SFT+RL rows of Table 14.

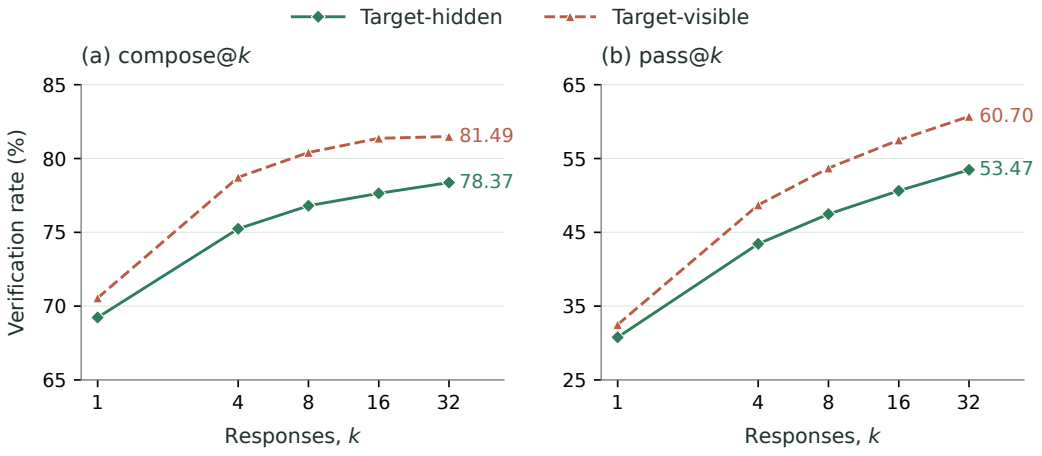

Figure 7: Effect of target visibility on a fixed target-hidden-trained Qwen3-8B SFT+RL checkpoint across all 832 programs. Each panel compares target-hidden (dashed squares) and target-visible (solid circles) inference at the same response budgets: (a) compose@k; (b) pass@k, both using the first k saved responses. The horizontal axes use a base-2 logarithmic scale; the panels use different vertical ranges. Visible compose rates are computed from exact success counts, while visible pass rates were reported to one decimal place. Neither condition uses iterative verifier-feedback refinement.

Figure 7 shows that target visibility improves both whole-workload metrics at every tested budget, even though the model was trained without targets. For $k \in \{4, 8, 16, 32\}$, compose@k increases by 3.12–3.73 percentage points, corresponding to a net increase of 26–31 verified programs. At k=1, the gain is 1.32 points (11 programs). The pass@k gain grows from approximately 1.72 points at k=1 to 7.23 points at k=32. Visible compose@4 (78.73%) also slightly exceeds hidden compose@32 (78.37%), demonstrating a practical effect on the response budget needed to reach this verification rate. These are descriptive results from the reported runs; aggregate rates alone do not establish statistical significance. Both metrics use the same saved prefix within each condition. As discussed in Appendix K, the visibility comparisons remain subject to sampling variance from the prefixes generated under each condition.

Table 15: CRAFT with target-visible (target-vision) and target-hidden inference using the same Qwen3-8B SFT+RL checkpoint, by benchmark stratum and sampling budget. Each cell reports a percentage; All / 832 is the whole-workload result. Visible compose rates are computed from exact success counts; visible pass rates retain the reported one-decimal precision. Hidden All results match Table 14.

		Linear / 316		NLA / 50		Loopy / 466		All / 832	
Configuration	k	pass	comp	pass	comp	pass	comp	pass	comp
CRAFT (target-hidden)	1	32.77	75.00	2.78	14.00	32.43	71.24	30.78	69.23
	4	46.39	79.75	3.83	14.00	45.69	78.76	43.44	75.24
	8	51.03	80.70	3.99	14.00	49.72	80.90	47.47	76.80
	16	54.65	80.70	4.00	14.00	52.92	82.40	50.63	77.64
	32	57.78	81.33	4.00	14.00	55.85	83.26	53.47	78.37
CRAFT (target-vision)	1	34.4	73.42	4.4	10.00	34.2	75.11	32.5	70.55
	4	51.5	80.70	5.9	14.00	51.4	84.33	48.7	78.73
	8	56.7	82.59	6.0	14.00	56.8	86.05	53.7	80.41
	16	60.7	83.23	6.0	14.00	60.9	87.34	57.5	81.37
	32	64.2	83.23	6.0	14.00	64.2	87.55	60.7	81.49

Table 15 shows that the composition gains vary by stratum. Loopy improves at every budget, by 3.87–5.57 percentage points. Linear improves for $k \geq 4$, but its compose@1 falls from 75.00% to 73.42%. NLA compose@1 falls from 14.00% to 10.00%, while both conditions reach 14.00% for $k \geq 4$. The whole-workload composition gains therefore do not imply a uniform benefit across strata.

Implications for feedback-based refinement. This control shows that exposing the target alone makes the evaluated tasks empirically easier for the fixed checkpoint. Target visibility also enables refinement to use failed target obligations to guide subsequent proposals, providing a source of target-directed feedback unavailable in our target-hidden training and inference protocol. Such feedback could further increase the performance gap, but this comparison does not measure that additional effect. Quantifying it requires a separate refinement experiment with matched model-call and verifier budgets. The present results therefore support treating target visibility and access to target-directed refinement as distinct protocol choices when comparing verification rates.

J.5 Clause-Cap Ablation

We vary the per-response clause cap while keeping the saved Qwen3-8B response pools and all other evaluation settings fixed. For cap c, only the first c parsed clauses from each response are admitted to standalone verification or pooled filtering. Table 16 therefore isolates how the number of available clauses changes the tradeoff between pass and composition at the Bare and SFT stages.

Table 16: Qwen3-8B clause-cap ablation (%). Each column uses the same saved response pools while varying the maximum number of admitted clauses per response.

	pass@1		compose@1		compose@8	
Cap	Bare	SFT	Bare	SFT	Bare	SFT
1	12.16	19.59	11.06	19.23	17.67	35.10
2	9.45	27.31	17.31	30.89	25.24	49.40
4	8.20	41.41	27.76	45.67	40.75	67.07
8	7.04	44.53	35.46	57.09	52.40	73.56
16	6.56	42.80	39.42	64.90	54.21	76.56

Table 17: Structural statistics for the Qwen3-8B clause-cap sweep. Clauses is the mean number of admitted clauses per response; Unique@8 is the mean number of canonical clause keys in the first eight responses; Survivors@8 is the mean number retained after filtering their pooled union.

Model	Statistic	Cap 1	Cap 2	Cap 4	Cap 8	Cap 16
Bare	Clauses	1.00	1.98	3.88	7.24	12.37
	Unique@8	2.72	5.50	11.67	24.30	41.69
	Survivors@8	1.42	2.49	5.37	11.33	19.91
SFT	Clauses	1.00	2.00	3.96	7.52	12.73
	Unique@8	3.37	6.69	14.26	30.14	56.07
	Survivors@8	2.92	5.71	12.10	25.27	46.35

For Bare Qwen3-8B, increasing the cap from 1 to 16 decreases pass@1 monotonically from 12.16% to 6.56%, while compose@1 rises from 11.06% to 39.42% and compose@8 from 17.67% to 54.21%. More clauses increase the chance that an unfiltered response contains a failure, but also enlarge the candidate set from which pooled filtering can retain useful constraints.

J.6 Ablation Results

Reward configuration. The reported RL runs apply the per-response admission cap without penalizing clauses beyond that cap, consistently with the reward definition above.

The reward variants use different correctness gates for Equation 10. *Binary Inductiveness* assigns one iff the processed response is nonempty and every clause survives Houdini. *Whole-Rollout Strength* pays the strength score $\mathrm{cov}(V_i)$ only when every processed clause survives,

$$r_i^{\mathrm{whole}} = \mathbf{1}[V_i = \mathrm{Unique}(\mathrm{First}_{m_{\max}}(\mathrm{parts}(y_i)))]\frac{|R_i|}{|\mathcal{N}|}.$$

Full decompose credit removes this all-or-nothing indicator and scores each response by the coverage of its surviving V_i. All three variants admit at most $m_{\max}$ clauses per response. We refer to them as Binary, Whole-rollout, and Full decompose credit in figures and prose. The first block trains all three variants from Qwen3-8B Bare; the remaining blocks report the available SFT-initialized controls. Within each block, variants share the training pool and optimization and inference settings.

Table 18: Reward-function ablation on all 832 programs (%). The first block trains Qwen3-8B from the Bare initialization and gives the exact numbers for Figure 4; the remaining blocks train from the SFT initialization. Binary uses binary inductiveness, Full decompose credit rewards surviving clauses, and Whole-rollout uses response-level Whole-rollout Strength. Each block lists the untrained checkpoint of its own initialization as the reference row; bold marks the best result per column among trained variants within each block. A dash means the control was not reported, not zero.

	pass@k					compose@k				
Variant	1	4	8	16	32	1	4	8	16	32
Qwen3-8B, trained from Bare										
Bare (untrained reference)	6.43	13.24	17.33	21.62	25.96	37.86	50.60	54.21	56.37	57.81
Binary Inductiveness	3.78	6.21	7.72	9.40	11.26	7.21	12.26	16.71	20.31	23.32
Whole-Rollout Strength	**16.67**	**19.17**	**20.03**	**20.90**	21.92	18.87	19.95	20.31	21.27	22.72
Full decompose credit	6.80	13.41	16.80	20.17	**23.44**	**57.93**	**66.95**	**70.43**	**72.60**	**73.20**
Qwen3-4B, trained from SFT										
SFT (untrained reference)	40.43	55.23	59.99	64.09	67.74	62.62	73.44	76.20	77.28	77.88
Whole-rollout Strength	**42.72**	**48.08**	**50.02**	51.81	53.54	46.15	51.08	54.69	56.61	58.77
Full decompose credit	33.40	46.10	49.90	**53.30**	**56.50**	**66.10**	**72.50**	**74.50**	**75.10**	**75.60**
Qwen3-8B, trained from SFT										
SFT (untrained reference)	41.93	56.82	61.11	64.68	67.98	63.90	73.80	76.20	77.30	77.90
Binary Inductiveness	13.90	16.40	17.60	18.60	19.50	13.70	16.10	17.90	18.80	19.70
Whole-rollout Strength	–	–	–	–	–	–	–	–	–	–
Full decompose credit	**30.78**	**43.44**	**47.47**	**50.63**	**53.47**	**69.23**	**75.24**	**76.80**	**77.64**	**78.37**
Qwen3-14B, trained from SFT										
SFT (untrained reference)	45.66	59.14	63.50	67.34	70.57	66.35	76.68	78.37	79.21	79.33
Whole-rollout Strength	–	–	–	–	–	–	–	–	–	–
Full decompose credit	**40.70**	**52.90**	**56.80**	**60.10**	**63.10**	**70.80**	**75.70**	**76.80**	**77.30**	**77.40**
Llama 3.1-8B, trained from SFT										
SFT (untrained reference)	39.24	52.59	56.86	60.47	63.70	61.30	70.31	73.08	74.04	74.40
Whole-rollout Strength	**44.30**	**52.60**	**55.90**	**58.90**	**61.70**	54.00	62.90	66.10	68.20	70.00
Full decompose credit	33.70	45.83	49.64	53.01	56.39	**63.70**	**70.70**	**72.70**	**73.10**	**73.20**

J.7 Alignment between Negative Coverage and Final Verification

This diagnostic asks whether the target-independent score distinguishes invariant sets that pass final verification from those that do not. We evaluate on the 832-program benchmark: each candidate set is scored by its structured relation/post-exit negative coverage, computed without $\mathcal{G}$, and labeled by the Frama-C/WP verdict after the untouched target is restored. Targets are used only to obtain evaluation labels, never to compute the score. Candidate sets are the archived outputs of the target-hidden GPT-5-nano pipelines used in the tool comparison (Table 20). The analysis covers the 691 programs that admit both verifier-accepted and rejected candidate sets—5,711 candidate sets in total—since per-program ranking requires both classes; the remaining 141 programs are either target-trivial or have no verified candidate. Each program contributes one verifier-accepted reference set, so the positive class is represented independently of the generation pipeline. AUROC is computed within each program and macro-averaged across programs; the 95% interval uses 5,000 bootstrap replicates over programs.

Population	Programs	Candidates	AUROC
Linear	248	2,173	0.881
NLA	49	439	0.830
Loopy	394	3,099	0.819
All	691	5,711	0.842

The overall AUROC is 0.842, with a program-level bootstrap 95% confidence interval of $[0.826, 0.858]$. Across all three categories, verifier-accepted invariant sets tend to reject more target-independent negative traces than rejected sets for the same program. This supports alignment between

the coverage score and the final verification objective; it does not imply that coverage alone guarantees verification or that the generator will discover a sufficient invariant.

This population consists of archived tool outputs and verified references, not response groups sampled from the current RL policy. Predictive alignment on these candidate sets does not by itself establish informative within-group reward differences during training.

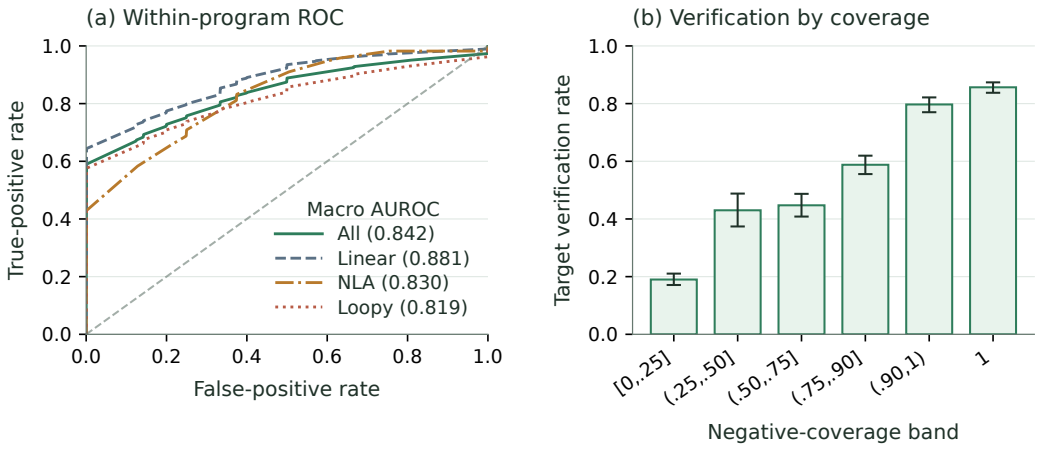

Figure 8: Predictive validity of target-independent negative coverage on the 5,711 evaluated candidate sets. (a) Within-program macro ROC curves for restored-target verification; numbers in parentheses are the macro AUROCs reported in the table above. (b) Target verification rate by coverage band, with Wilson 95% intervals.

J.8 Cross-Model Results

Table 19: Complete reasoning-disabled cross-model results by benchmark stratum and sampling budget. Each cell reports a percentage; All / 832 is the whole-workload result. Rows for each model come from one archived rollout pool of 10 responses per task. Both pass@k and compose@k evaluate the same first k saved responses. Bold marks the best value in each column at a fixed budget. GPT-5-mini is excluded because it has no reasoning-disabled setting.

		Linear / 316		NLA / 50		Loopy / 466		All / 832	
Model	k	pass	comp	pass	comp	pass	comp	pass	comp
GPT-5-nano	1	3.77	34.81	0.40	22.00	3.80	34.55	3.58	33.89
	4	10.79	48.10	1.60	30.00	11.48	51.72	10.62	49.04
	8	15.84	55.38	3.20	44.00	17.58	56.87	16.05	55.53
GPT-5	1	**35.06**	52.22	5.80	36.00	33.78	52.36	32.58	51.32
	4	**47.73**	64.87	8.74	52.00	**50.67**	67.38	**47.03**	65.50
	8	**51.93**	70.25	**9.60**	66.00	**56.13**	73.82	**51.74**	72.00
Claude Sonnet-4.6	1	34.49	**55.38**	**7.80**	**44.00**	**40.67**	**59.44**	**36.35**	**56.97**
	4	40.32	62.97	**8.80**	52.00	47.27	66.31	42.32	64.18
	8	42.87	66.46	**9.60**	64.00	50.12	67.60	44.93	66.95
DeepSeek V4-Flash	1	28.67	52.22	5.00	**44.00**	30.15	51.72	28.08	51.44
	4	42.35	**66.46**	7.52	**60.00**	47.73	**69.96**	43.27	**68.03**
	8	46.64	**71.52**	8.00	**72.00**	53.59	**76.61**	48.21	**74.40**

Pass@k and compose@k evaluate the same first k archived responses per task for each model. Pass@k checks whether any response in that prefix succeeds alone; compose@k filters and composes its clause union.

J.9 Complete Comparison with Specialized Tools

Table 20: Complete target-hidden tool comparison underlying Figure 3: the reasoning-disabled rows alongside the archived medium-reasoning rerun; Daikon makes no model call and is listed once. Each cell reports the verification rate for Linear, NLA, Loopy, or all 832 programs. Medium-effort results are available only for the single-response measurements. The reasoning-disabled compose@1, compose@4, and compose@8 rows are prefix compositions of the same archived ten-rollout pool.

Method	Reason.	Linear	NLA	Loopy	All	Tokens/task	Time/task
AutoSpec	none	47.78%	6.00%	42.27%	42.19%	2,181.72	27.34 s
AutoSpec	medium	65.82%	8.00%	64.81%	61.78%	25,154.54	54.77 s
SESpec	none	28.80%	4.00%	33.05%	29.69%	22,081.61	68.50 s
SESpec	medium	56.96%	6.00%	49.57%	49.76%	44,807.05	117.31 s
Clause2Inv	none	20.89%	0.00%	0.00%	7.93%	1,056.25	8.41 s
Clause2Inv	medium	19.94%	2.00%	0.00%	7.69%	385.80	29.04 s
Daikon	—	23.10%	0.00%	21.67%	20.91%	0	4.89 s
Loopy	none	20.25%	0.00%	19.74%	18.75%	14,513.37	147.16 s
Loopy	medium	72.78%	12.00%	70.82%	68.03%	111,973.15	381.01 s
Naive	none	15.19%	2.00%	18.88%	16.47%	225.95	2.98 s
Naive	medium	48.73%	8.00%	47.42%	45.55%	4,493.37	28.87 s
CRAFT (pass@1)	none	2.53%	0.00%	4.72%	3.61%	1,135.83	7.86 s
CRAFT (pass@1)	medium	61.08%	14.00%	54.08%	54.33%	6,928.30	53.52 s
CRAFT (compose@1)	none	34.81%	22.00%	34.55%	33.89%	1,136.82	24.52 s
CRAFT (compose@1)	medium	64.24%	14.00%	63.09%	60.58%	6,928.30	66.60 s
CRAFT (compose@4)	none	48.10%	30.00%	51.72%	49.04%	1,905.10	46.40 s
CRAFT (compose@8)	none	55.38%	44.00%	56.87%	55.53%	2,929.47	69.99 s

The main text reports each method's reasoning-disabled row. The shared model-call parameters for the reasoning-disabled comparison are summarized separately in Table 21.

Table 21: Sampling parameters used in the reasoning-disabled target-hidden tool comparison.

Method	Sampling budget / task	Temperature / top-p	Completion cap	Reasoning
AutoSpec	8	1.0 / 1.0	8,192	none
SESpec	1 initial + up to 3 refinements per phase	1.0 / 1.0	8,192	none
Clause2Inv	adaptive; 1 per call	1.0 / 1.0	8,192	none
Daikon	no model call	–	–	–
Naive	1	1.0 / 1.0	8,192	none
CRAFT	1, 4, or 8	1.0 / 1.0	8,192	none
Loopy	15 initial + up to 7 repairs	0.7 / 1.0	8,192	none

Token totals are means over tasks with at least one model call; accuracy always keeps the 832-task denominator. AutoSpec makes one request that returns eight proposals; its reasoning-disabled token count is the native estimate for that complete request. Four failed tasks have no token record, so its reported token mean uses the remaining 828 tasks.

We close with method-specific notes under target hiding.

AutoSpec. AutoSpec pools eight parallel proposals in its native orchestration. The reported result is produced by this multi-proposal pipeline.

SESpec. SESpec does not pool independently sampled responses as CRAFT does; its additional calls iteratively repair a running candidate. With reasoning disabled, it reaches 29.69%, close to CRAFT compose@1 at 33.89%, while using 22,081.61 versus 1,136.82 mean tokens per task (19.4×).

Clause2Inv. Clause2Inv normally constructs false-state counterexamples from the postcondition. Under target hiding, this source of counterexamples is unavailable. Its transition verification condi-

tions are also unavailable for the 466 Loopy programs; these instances remain failures in the common denominator.

Loopy. Loopy's initial responses contain textual loop-invariant annotations in 98.8% of cases, and its native extractor passes 1.75% of initial responses to Houdini. These rates measure the interface between generated text and the clauses accepted by its downstream pipeline.

K Detailed Limitations

Sampling variance of prefix evaluation. Both pass@k and compose@k use the same first k saved responses per task. We evaluate one prefix at each budget without averaging over alternative size-k subsets or independent generation runs. This keeps composition evaluation tractable, since each alternative subset requires a separate Houdini filtering and final verification run. Sharing the prefix controls which responses the two methods receive, but does not remove sampling variance: changing the generation seed or saved order can change which responses enter the prefix and hence both verification rates and their gap. At $k{=}1$, each task contributes only one sampled response to each metric. Small differences and apparent saturation can therefore reflect the particular sampled prefixes. Moreover, prefixes are nested across budgets, so points on a curve are correlated rather than independent replications. The reported curves characterize the saved runs; they do not establish statistical significance or equal variability for the two metrics. A stronger evaluation would repeat generation or sample multiple matched subsets for both methods, and report uncertainty for the paired differences.

Program scope. Our implementation targets numerical programs with one scalar-integer loop. The evaluation excludes nested loops and invariants over arrays, heaps, or recursively defined predicates. The negative sampler covers scalar arithmetic relations and exit boundaries (Liu et al., 2024a).

Inference engine. Local base and trained checkpoints are served with vLLM. In preliminary comparisons under nominally matched decoding settings, these checkpoints had lower verification rates with vLLM than with the other evaluated backends. API models retain their provider serving interfaces and chat templates. The reported decoding and reasoning controls are applied to each backend.

Verifier and specification language. Our filtering and all final judgments use Frama-C/WP, and candidates use ACSL. Parser behavior, generated obligations, prover integration, and syntax affect which clauses survive. The reported results characterize this verifier, specification language, and source language.

Finite candidate-pool approximation. Before Houdini, the pipeline implementation applies a lightweight AST-based filter to parse, normalize, and deduplicate candidate clauses. Clauses outside its supported ACSL subset are omitted. Each response contributes at most $m_{\max}$ clauses before global pooling; later clauses in that response are truncated, while no separate cap is applied to the pooled union.

Feedback regime. The evaluation covers target-independent training and finite-budget composition without iterative verifier-feedback optimization. In pilot runs, models reached inductive but weak invariants: the invariants passed induction, while the hidden goals left the verifier without a target-directed signal for the missing behavioral strength. The marginal gain from refinement decreased as the initial rollout pool grew.

Training variance. We did not repeat training and inference multiple times for every experimental configuration.

L WHY THINKING IS DISABLED FOR LOCAL MODELS

The reasoning-disabled API experiments use compatible endpoint controls; Claude Sonnet disables adaptive thinking with a zero thinking budget. Local checkpoints use reasoning-disabled decoding during synthetic-data construction, training rollouts, and evaluation.

Explicit chain-of-thought adds reasoning tokens to every rollout, with total generation scaling with the sampling budget. The required output is a short, machine-parseable set of ACSL clauses. These clauses admit clause-level filtering and cross-rollout composition: Houdini can remove a failing clause, retain its siblings, and merge survivors from different rollouts. Reasoning traces are generated and consumed as whole sequences and are not included in this clause-level composition operator.

M POOLED FILTERING AND PROVENANCE CREDIT

Equations (29)–(30) define the common order: merge the clauses, filter the union once, and assign each survivor back to its proposers. RL, SFT, and inference use this pooled order.

Pre-filtering and post-filtering. Let $\mathsf{F}(D) = \mathrm{Filter}_{\mathcal{L}}(D)$ and

$$\mathsf{U}(D_1, \ldots, D_k) = \mathrm{Unique}(D_1 \| \cdots \| D_k).$$

The two placements differ only in whether filtering occurs before or after the response pools are combined:

$$C_{\mathrm{pre}} = \mathsf{U}\big(\mathsf{F}(A_1), \ldots, \mathsf{F}(A_k)\big), \tag{40}$$

$$C_{\mathrm{post}} = \mathsf{F}\big(\mathsf{U}(A_1, \ldots, A_k)\big). \tag{41}$$

Thus pre-filtering makes k independent verifier calls and discards a clause before it can interact with clauses from another response. Post-filtering makes one call on the pooled set, allowing cross-response support before deletion (Proposition 4).

The corresponding provenance assigned to response i is

$$\begin{aligned} V_i^{\mathrm{pre}} &= \mathsf{F}(A_i), \\ V_i^{\mathrm{post}} &= \langle c \in A_i \mid \mathrm{key}(c) \in \mathrm{keys}(C_{\mathrm{post}}) \rangle \,. \end{aligned} \tag{42}$$

Consequently, the post-filter reward uses $R_i = \mathrm{rej}_{\mathcal{N}}(V_i^{\mathrm{post}})$, so credit is computed from clauses that survive in the same pooled context used at inference.

For ideal Houdini, Equation (3), proved in Appendix B, establishes that post-filtering retains all pre-filtered clauses. In the resource-bounded implementation, larger pooled obligations can time out, so this logical advantage need not translate into higher measured verification rates. We examine that accuracy–cost tradeoff below.

Table 22 reports both placements on the 832-program benchmark with Qwen3-8B, evaluating saved prefixes up to 32 rollouts. Under the 300-second cap, post-filtering reaches 58.7 at k=32, compared with 64.4 for pre-filtering. With the 900-second cap, the corresponding result is 64.2. Whole-workload costs are 113, 127, and 1,472 CPU-hours for short-cap post-filtering, long-cap post-filtering, and pre-filtering, respectively.

Table 22: Pre-filtering and post-filtering for invariant verification on the 832-program benchmark (Qwen3-8B, saved prefixes up to 32 rollouts; compose@k in %). CPU-h is the whole-workload total.

Method	Subset	@1	@4	@8	@16	@32	CPU-h
Post-filter (300 s)	All	40.4	52.8	56.0	58.1	58.7	113
	Loopy	43.3	56.9	60.9	63.1	63.7	
	Linear	41.1	53.8	56.3	57.9	58.5	
	NLA	8.0	8.0	8.0	12.0	12.0	
Post-filter (900 s)	All	40.4	53.4	58.4	62.5	64.2	127
	Loopy	43.3	57.7	63.3	68.7	70.2	
	Linear	41.1	54.1	59.2	61.4	63.3	
	NLA	8.0	8.0	8.0	12.0	14.0	
Pre-filter	All	40.4	52.8	57.9	62.1	64.4	1472
	Loopy	43.3	57.3	62.9	68.0	71.0	
	Linear	41.1	53.2	58.5	61.4	63.0	
	NLA	8.0	8.0	8.0	12.0	12.0	